HAIMAN'S CONJECTURE AND SPRINGER'S REPRESENTATIONS

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ABSTRACT. For any connected complex reductive group G and element z of its Weyl group W, we use work of Lusztig and Abreu–Nigro to compute the graded W-character of the intersection cohomology of any closed Lusztig variety for z over the regular semisimple locus of G. We relate the resulting formula to unipotent Lusztig varieties, giving a new geometric model for unicellular LLT polynomials. We then consider Laurent polynomials $\alpha_{\psi,G}^z$ indexed by irreducible characters ψ, encoding how our formula decomposes into ungraded characters arising from the Springer theory of G. From evidence in low rank, we conjecture that if ψ is inflated from type A in a particular way, then the nonzero coefficients of $\alpha_{\psi,G}^z$ are positive and unimodal. This offers an answer to a 1993 question of Haiman about generalizing a conjecture he posed for symmetric groups. We also prove that the matrix formed by the $\alpha_{\psi,G}^z$ is partially triangular, and that their positivity and unimodality properties are stable under inclusions of Levi subgroups.

1. Conjectures and Results

1.1. **Motivation.** The subject of this paper is how to generalize a 1993 conjecture of Haiman from symmetric groups to Weyl groups.

For any finite Coxeter group W, let H_W be the Hecke algebra of W over $\mathbf{Z}[\mathsf{v}^{\pm 1}]$ with standard basis $\{\delta_w\}_{w \in W}$. We fix a system of simple reflections $S \subseteq W$ and normalize the elements δ_w so that H_W, as an algebra, is generated by the set $\{\delta_s\}_{s \in S}$ modulo the relations $\delta_s^2 = (\mathsf{v}^2 - 1)\delta_s + \mathsf{v}^2$ and so-called braid relations. Let $\{c_w\}_{w \in W}$ be the Kazhdan–Lusztig basis of H_W denoted $\{C'_w\}_{w \in W}$ in [KL79]. It is invariant under the ring anti-automorphism sending $\mathsf{v} \mapsto \mathsf{v}^{-1}$ and $\delta_s \mapsto \delta_s^{-1}$ for all $s \in S$. For instance, if $s \in S$, then $c_s = \mathsf{v}^{-1}(1 + \delta_s) = \mathsf{v}(1 + \delta_s^{-1})$.

Let $\mathrm{Irr}(W)$ be the set of irreducible characters of W. The representation theory of H_W deforms that of W, just as H_W itself deforms the group ring $\mathbf{Z}W$. In particular, each class function χ on W deforms to a $\mathbf{Z}[\mathsf{v}^{\pm 1}]$-linear function χ_{v} on H_W that remains a trace: *i.e.*, vanishes on commutators. If W is a Weyl group and $\chi \in \mathrm{Irr}(W)$, then χ takes values in $\mathbf{Z}$, and χ_{v} takes values in $\mathbf{Z}[\mathsf{v}^{\pm 1}]$.

When W is the symmetric group S_n, the theory of cell modules shows that $\chi_{\mathsf{v}}(c_z) \in \mathbf{Z}_{\geq 0}[\mathsf{v}^{\pm 1}]$ for all $z \in W$ and $\chi \in \mathrm{Irr}(W)$, a positivity statement that already fails when W is the Weyl group of type BC_2. Haiman conjectured a much stronger positivity. To state it, first recall that the elements of $\mathrm{Irr}(S_n)$ are indexed by integer partitions of n. For any $\lambda \vdash n$, let $\chi_\lambda \in \mathrm{Irr}(S_n)$ be the corresponding irreducible character. Next, recall that the Frobenius characteristic map [Mac95, §1.7] is a linear bijection between class functions on S_n and symmetric functions of degree n (say, in n variables), taking χ_λ to the Schur function indexed by λ.

For any $\mu \vdash n$, let $mon_\mu \colon S_n \to \mathbf{Z}$ be the class function whose Frobenius characteristic is the *monomial* symmetric function indexed by μ. Then

$$\chi_\lambda = \sum_{\mu \vdash n} K_{\lambda,\mu} mon_\mu,$$

where $K_{\lambda,\mu} \in \mathbf{Z}_{\geq 0}$ is the Kostka number counting semistandard Young tableaux of shape λ and weight μ. For any $z \in S_n$, let

$$\alpha_\mu^z(\mathsf{v}) = mon_{\mu,\mathsf{v}}(c_z) \in \mathbf{Z}[\mathsf{v}^{\pm 1}].$$

What follows is Conjecture 2.1 in [Hai93].

Conjecture A (Haiman 1993)**.** *For any $z \in S_n$ and $\mu \vdash n$, the nonzero coefficients of α_μ^z are all positive and form a unimodal sequence.*

Haiman points out that the coefficients of $\alpha_\mu^z(\mathsf{v})$ are at least palindromic, due to the invariance of c_z under the anti-automorphism sending δ_s, v to $\delta_s^{-1}, \mathsf{v}^{-1}$.

Further motivation for Conjecture A appeared in [CHSS16], where Clearman, Hyatt, Shelton, and Skandera showed that the positivity of $mon_{\mu,\mathsf{v}}(c_z)$ would subsume the Shareshian–Wachs conjecture: the e-positivity of the chromatic quasisymmetric functions of unit interval graphs [SW16]. Thus it would also subsume the Stanley–Stembridge conjecture [Sta95, SS93] recently proved in [Hik25, GMR+25].

To explain the connection, let $ind_\mu \colon S_n \to \mathbf{Z}$ be the class function whose Frobenius characteristic is the homogeneous symmetric function indexed by μ. Equivalently, ind_μ is the character of the representation of S_n induced from the parabolic subgroup $S_\mu \simeq S_{\mu_1} \times S_{\mu_2} \times \dots$ Then

$$ind_\mu = \sum_{\lambda \vdash n} K_{\lambda,\mu} \chi_\lambda.$$

So the Laurent polynomials $\alpha_\mu^z(\mathsf{v})$ are uniquely determined by requiring that

$$\sum_{\lambda \vdash n} \chi_{\lambda,\mathsf{v}}(c_z)\chi_\lambda = \sum_{\mu \vdash n} \alpha_\mu^z(\mathsf{v}) ind_\mu \tag{1.1}$$

as functions on S_n.

Each unit interval graph on n vertices corresponds to some permutation $z \in S_n$ that is *312-avoiding*,[1] meaning no indices $1 \leq i < j < k \leq n$ exist for which $z(j) < z(k) < z(i)$. The authors of [CHSS16] show that up to a sign twist and power of v, the Frobenius characteristic of the left-hand side of (1.1) for this z is the chromatic quasisymmetric function for the graph. In this way, Conjecture A would imply the positivity conjectured by Shareshian–Wachs.

For further details about unit interval graphs and their chromatic quasisymmetric functions, we defer to the references above.

[1] In Haiman's terminology, *codominant*.

1.2. **Lusztig Varieties.** At the end of [Hai93, §2], Haiman writes:

> There is probably an analog of Conjecture 2.1 for other Coxeter groups or at least Weyl groups. At present, however, it is not apparent what the analog of a monomial character should be.

Though we still lack a meaningful generalization of the characters mon_μ beyond S_n, we will present a way to generalize Conjecture A to finite Weyl groups, using the work of Springer and Lusztig in algebraic geometry.

Already in [SW16], Shareshian–Wachs proposed an algebro-geometric formula[2] for the chromatic quasisymmetric function of any unit interval graph, in terms of an S_n-action on the singular cohomology of a complex algebraic variety associated with the graph. In Lie-theoretic language, these varieties are Hessenberg varieties for the group GL_n, depending not only on the graph, but on a regular semisimple element in the Lie algebra $\mathfrak{gl}_n$. The formula was proved by Brosnan–Chow [BC18]. In particular, Brosnan–Chow showed how to construct the S_n-action from the monodromy of a family of Hessenberg varieties over the regular semisimple locus of $\mathfrak{gl}_n$.

Hessenberg varieties can be generalized to any reductive algebraic group G, by replacing unit interval graphs with so-called Hessenberg subspaces of the Lie algebra of G. This suggests a generalization of the e-positivity conjecture in terms of Hessenberg varieties, replacing S_n with the Weyl group of G. More daringly, it suggests a generalization of Conjecture A in terms of varieties associated with elements of the Weyl group.

In [AN25], Abreu–Nigro showed that the left-hand side of (1.1) records a monodromy representation on the *intersection* cohomology of a variety $\bar{Y}_{z,g}$ built from z and an arbitrary choice of regular semisimple element $g \in \mathrm{GL}_n(\mathbf{C})$. This variety is now called a Lusztig variety, as it was first introduced by Lusztig in [Lus79]. Explicitly, if $\mathcal{B}$ denotes the flag variety of G, and W denotes its Weyl group, then for any $z \in W$ and $g \in G(\mathbf{C})$, the *closed Lusztig variety* $\bar{Y}_{z,g}$ is the Zariski closure in $\mathcal{B}$ of the *open Lusztig variety*

$$Y_{z,g} := \{B \in \mathcal{B} \mid B \xrightarrow{z} g^{-1}Bg\},$$

where $B \xrightarrow{z} B'$ indicates that a pair of Borels (B, B') is in relative position z. Intersection cohomology is needed because $\bar{Y}_{z,g}$ may be singular. Abreu–Nigro's result harmonizes with recent work of Brosnan–Hong–Lee [BHL25], showing that when a (smooth) element $z \in W$ corresponds to a Hessenberg subspace $\mathfrak{H}$ of the Lie algebra of G, the W-representation arising from a Lusztig variety for z matches that arising from a Hessenberg variety for $\mathfrak{H}$.

In this paper, we will use étale cohomology theories with ℓ-adic coefficients. Let IH^* denote intersection cohomology. For a general connected complex reductive algebraic group G, we write G^{rs} to denote its regular semisimple locus, a Zariski dense open subset. We are led to seek a generalization of Conjecture A that replaces S_n with its Weyl group W and the left-hand side of (1.1) with $\mathrm{IH}^*(\bar{Y}_{z,g})$ for $g \in G^{rs}(\mathbf{C})$.

[2]Another statement known as the Shareshian–Wachs conjecture.

It turns out that $\mathrm{IH}^i(\bar{Y}_{z,g})$ vanishes for odd i and its top degree is $\ell(z)$, the Bruhat length of z. Moreover, by a result of Abreu–Nigro, the graded representation of W on $\mathrm{IH}^*(\bar{Y}_{z,g})$ is constant up to isomorphism as g varies in $G^{rs}(\mathbf{C})$. Our first theorem is a purely algebraic formula for this representation in terms of a pairing $\{-,-\}\colon \mathrm{Irr}(W) \times \mathrm{Irr}(W) \to \mathbf{Q}$ called the *(truncated) exotic Fourier transform*, originating in Lusztig's work on characters of finite reductive groups [Lus84]. We review its definition in Appendix A. For W of classical type, Lusztig gives a purely combinatorial algorithm to calculate $\{-,-\}$.

For any $z \in W$, let $\tau_{z,\mathsf{v}}\colon W \to \mathbf{Z}[\mathsf{v}^{\pm 1}]$ be defined by

$$\tau_{z,\mathsf{v}} = \sum_{\chi,\psi \in \mathrm{Irr}(W)} \{\chi,\psi\}\psi_{\mathsf{v}}(c_z)\chi.$$

Note that $\{-,-\}$ and hence $\tau_{z,\mathsf{v}}$ only depend on W, not on G. In Section 3, we use the work of Lusztig and Abreu–Nigro to prove the following formula.

Theorem 1.1. *Fix a connected complex reductive group G with Weyl group W. Then for any $z \in W$ and $g \in G^{rs}(\mathbf{C})$ and $w \in W$, we have*

$$\sum_i \mathsf{v}^{2i}\,\mathrm{tr}(w \mid \mathrm{IH}^{2i}(\bar{Y}_{z,g})) = \mathsf{v}^{\ell(z)}\tau_{z,\mathsf{v}}(w),$$

where the W-action on the intersection cohomology arises from the monodromy action of the fundamental group of G^{rs} based at g.

Example 1.2. In type A, where $W = S_n$, the exotic Fourier transform is simply the pairing where $\{\chi,\chi\} = 1$ and $\{\chi,\psi\} = 0$ for $\chi \neq \psi$. So here, $\tau_{z,\mathsf{v}}$ recovers the left-hand side of (1.1), and Theorem 1.1 recovers [AN25, Theorem 1.5].

The linear function on H_W that sends $c_z \mapsto \tau_{z,\mathsf{v}}$, being a linear combination of the characters ψ_{v}, may be viewed as a trace on H_W. It appeared in [Tri21], in relation to Lusztig varieties over the *unipotent* locus $G^{uni} \subseteq G$. Pursuing the viewpoint there yields a parallel interpretation of $\tau_{z,\mathsf{v}}$ in terms of G^{uni} rather than G^{rs}, as we now explain.

Recall that in Lusztig's work, the varieties $Y_{z,g}$, *resp.* $\bar{Y}_{z,g}$, are the fibers of maps $Y_z \to G$, *resp.* $\bar{Y}_z \to G$. For the identity element $z = e$, we have $\bar{Y}_e = Y_e$. The restricted map $Y_e^{uni} \to G^{uni}$ is a resolution of singularities of G^{uni} now called the *Springer resolution*. Springer discovered a remarkable action of W on the cohomologies of its fibers, not induced by an action on the fibers themselves. More precisely, Springer worked with an analogue for the Lie algebra of G, and over a finite field, not over $\mathbf{C}$. The Fourier–Deligne transform intertwines his action with a monodromy action over the regular semisimple locus of the Lie algebra.

Up to a sign twist, the action of W can also be constructed from convolution diagrams involving the *Steinberg variety*

$$\mathcal{Z} = Y_e^{uni} \times_{G^{uni}} Y_e^{uni}.$$

Convolution turns the Borel–Moore homology of $\mathcal{Z}$ into a graded algebra that acts on the cohomology of any Springer fiber. Motivated by [Tri21], we introduce

$$\bar{\mathcal{Z}}_z = Y_e^{uni} \times_G \bar{Y}_z. \tag{1.2}$$

The Borel–Moore homology $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{Z})$ then acts by convolution on the compactly-supported cohomology $\mathrm{H}_c^*(\bar{\mathcal{Z}}_z)$.

As in [Tri21], it is more convenient to work G-equivariantly. We have $\mathrm{H}_*^{\mathrm{BM},G}(\mathcal{Z}) \simeq \mathbf{C}W \otimes \mathrm{Sym}^*(\mathbb{X})$, where $\mathbb{X}$ denotes the character lattice in the root datum of G, on which W acts as a reflection group, and $\mathrm{Sym}^*(\mathbb{X})$ denotes the symmetric algebra generated by $\mathbb{X}$ in degree 2. Let $\mathbb{S}_{\mathsf{q}} : W \to \mathbf{Z}[\![\mathsf{q}]\!]^\times$ be the graded character

$$\mathbb{S}_{\mathsf{q}}(w) := \sum_i \mathsf{q}^i \operatorname{tr}(w \mid \mathrm{Sym}^i(\mathbb{X})) = \frac{1}{\det(1 - \mathsf{q}w \mid \mathbb{X})}.$$

With this notation, we can state a unipotent counterpart to Theorem 1.1. In what follows, recall that z is *rationally smooth* if and only if the Kazhdan–Lusztig polynomial of (e, z) is trivial [KL79].

Theorem 1.3. *Fix a connected complex reductive group G with Weyl group W. If $z \in W$ is rationally smooth, then for all $w \in W$, we have*

$$\sum_j (-\mathsf{v})^j \operatorname{tr}(w \mid \mathrm{H}^j_{c,G}(\bar{\mathcal{Z}}_z)) = (-1)^{\ell(w)} \mathsf{v}^{2\dim G + \ell(z)} \mathbb{S}_{\mathsf{v}^2}(w) \tau_{z,\mathsf{v}}(w),$$

where the W-action on the equivariant compactly-supported cohomology arises from convolution.

In Section 4, we prove this result, and sketch a generalization that applies to all z, not just rationally smooth z.

We mentioned earlier that the Shareshian–Wachs symmetric function for a unit interval graph is essentially the Frobenius characteristic of $\tau_{z,\mathsf{v}}$ for some 312-avoiding element $z \in S_n$. Proposition 3.1 of [Hai93] shows that if z is a 312-avoiding, then its Schubert variety is smooth, so z is rationally smooth.

When $\mathbb{X}$ has rank n, Frobenius characteristic transports the pointwise product with $\mathbb{S}_{\mathsf{q}}$ to the plethystic transform $X_i \mapsto X_i/(1-\mathsf{q})$ in the symmetric-function variables $X_1, X_2, \ldots$ The plethystic transforms of the Shareshian–Wachs functions appear in the work of Carlsson–Mellit on the shuffle conjecture, where they are called *characteristic functions of Dyck paths* [CM18]. As explained in [AP18], they are precisely the Lascoux–Leclerc–Thibon (LLT) polynomials for tuples of shifted partitions in which each partition has size ≤ 1. Hence, Theorem 1.3 provides a new algebro-geometric interpretation of these *unicellular* LLT polynomials.

Remark 1.4. In [Kat24], S. Kato geometrizes Shareshian–Wachs functions in terms of varieties over the affine Grassmannian of GL_n. It would be interesting to know whether the map $\bar{\mathcal{Z}}_z \to G^{uni}$, for 312-avoiding $z \in S_n$, is related to Kato's work: say, via Lusztig's embedding of the nilpotent cone into the affine Grassmannian.

1.3. **The Springer Correspondence.** Returning to Conjecture A, we next seek to generalize the characters ind_μ on the right-hand side of (1.1). It turns out that simply replacing them with the characters of W induced from trivial characters of parabolic subgroups does not work, because these are too few. Fortuitously, the work of Springer discussed above offers another generalization.

In what follows, for all $u \in G^{uni}(\mathbf{C})$, we replace the notation $Y_{e,u}$ with the more standard notation $\mathcal{B}_u$. Explicitly,

$$\mathcal{B}_u = \{B \in \mathcal{B} \mid u \in B\}.$$

A purity result shows that $\mathrm{H}^i(\mathcal{B}_u)$, like $\mathrm{IH}^*(\bar{Y}_{z,g})$, vanishes for odd i.

Let $C_u \subseteq G$ be the conjugacy class of u and $G_u \subseteq G$ its centralizer. Via the isomorphism of varieties $C_u \simeq G/G_u$, any representation of the component group $A_u := \pi_0(G_u)$ defines a local system on C_u. The cohomology of $\mathcal{B}_u$ decomposes into pieces indexed by such local systems. Indeed, the G-action on $\mathcal{B}$ restricts to a G_u-action on $\mathcal{B}_u$, which induces an A_u-action on $\mathrm{H}^*(\mathcal{B}_u)$. The actions of W and A_u on $\mathrm{H}^*(\mathcal{B}_u)$ centralize each other. For any $\kappa \in \mathrm{Irr}(A_u)$, we write $\mathrm{H}^*(\mathcal{B}_u)_\kappa$ to denote the multiplicity space of κ in $\mathrm{H}^*(\mathcal{B}_u)$.

The *Springer correspondence* states that each irreducible character $\psi \in \mathrm{Irr}(W)$ is afforded by the W-action on $\mathrm{H}^{2\dim \mathcal{B}_u}(\mathcal{B}_u)_\kappa$ for some unipotent u and $\kappa \in \mathrm{Irr}(A_u)$, and that the pair (u, κ) is uniquely determined up to conjugacy by ψ. For such ψ and (u, κ), let $spr_{\psi,G} = spr_{u,\kappa} \colon W \to \mathbf{Z}$ be defined by

$$spr_{\psi,G}(w) = spr_{u,\kappa}(w) := \sum_i \mathrm{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa). \tag{1.3}$$

These functions depend on the root system of G, though not on the isomorphism class of G itself. In type A, if G has connected center, then the groups A_u are trivial, and $spr_{\chi_\mu,G}$ recovers ind_μ. In other classical types, D. Kim gives a purely combinatorial formula for a coarser function, the character of the total cohomology $\mathrm{H}^*(\mathcal{B}_u)$, in terms of modified Kostka polynomials [Kim18]. In all types, the linear operator on class functions that sends $\psi \mapsto spr_{\psi,G}$ is unipotent, hence invertible.

For any $z \in W$, let $\alpha^z_{\psi,G}(\mathsf{v}) \in \mathbf{Z}[\mathsf{v}^{\pm 1}]$ be the (necessarily palindromic) Laurent polynomials indexed by $\psi \in \mathrm{Irr}(W)$ such that

$$\tau_{z,\mathsf{v}} = \sum_{\psi \in \mathrm{Irr}(W)} \alpha^z_{\psi,G}(\mathsf{v}) spr_{\psi,G} \tag{1.4}$$

as functions on W. It would be pleasant to generalize Conjecture A by generalizing Haiman's α^z_μ to our $\alpha^z_{\psi,G}$. Unfortunately, the nonzero coefficients of $\alpha^z_{\psi,G}$ can fail to be positive. In many cases, they are all negative, but in rare cases, they have mixed signs. They can also fail to form a unimodal sequence, even in absolute value.

Remark 1.5. A result from the character theory of finite reductive groups, the *orthogonality of Green functions*, implies that up to a polynomial denominator, the function $\mathsf{v}^{\ell(z)}\mathbb{S}_{\mathsf{v}^2} \cdot \tau_{z,\mathsf{v}}$ in Theorem 1.3 is a $\mathbf{Z}[\mathsf{v}^2]$-linear combination of the functions

$spr_{u,a,\mathsf{v}^2} \colon W \to \mathbf{Z}[\mathsf{v}^2]$ defined by

$$spr_{u,a,\mathsf{v}^2}(w) = \sum_i \mathsf{v}^{2i} \sum_{\kappa \in \mathrm{Irr}(A_u)} \kappa(a)\,\mathrm{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa),$$

as we run over $u \in G^{uni}(\mathbf{C})$ and $a \in A_u$. Elsewhere, we will discuss a geometric interpretation of the coefficients in this expansion, which superficially resembles (1.4). But so far, we do not know any direct relationship between the two expansions.

We introduce the following shorthand for the properties that interest us:

$(\triangle)^z_{\psi,G}$: the nonzero coefficients of $\alpha^z_{\psi,G}$ are unimodal in absolute value,

$(\pm)^z_{\psi,G}$: the nonzero coefficients of $\alpha^z_{\psi,G}$ are either all positive or all negative,

$(+)^z_{\psi,G}$: the nonzero coefficients of $\alpha^z_{\psi,G}$ are all positive.

The examples in Section 7 lead us to believe that as the rank of W grows, the elements z where $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ hold for all ψ, but not $(+)^z_{\psi,G}$, come to predominate in W. At the same time, we believe that the irreducible characters ψ where $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ hold for all z, but not $(+)^z_{\psi,G}$, come to predominate in $\mathrm{Irr}(W)$.

Question 1.6. Given $\omega \in \{\triangle, \pm, +\}$:

(I) What properties of $z \in W$ ensure that $(\omega)^z_{\psi,G}$ holds for all ψ?
(II) What properties of $\psi \in \mathrm{Irr}(W)$ ensure that $(\omega)^z_{\psi,G}$ holds for all z?

Remark 1.7. The truth of these properties does depend on the root system of G, not just on W: In the examples in Section 7, they fail for more pairs (z, ψ) in type C_n than in type B_n.

Remark 1.8. Suppose that G is isogenous to $G_1 \times G_2$. Let W_i be the Weyl group of G_i, so that $W \simeq W_1 \times W_2$. Then the pairing $\{-, -\}$ and the functions $\psi \mapsto spr_{\psi,G}$ are multiplicative with respect to the bijection $\mathrm{Irr}(W) \simeq \mathrm{Irr}(W_1) \times \mathrm{Irr}(W_2)$. We deduce that if $(\omega)^{z_1}_{\psi_1,G_1}$ and $(\omega)^{z_2}_{\psi_2,G_2}$ hold for some $z_i \in W_i$ and $\psi_i \in \mathrm{Irr}(W_i)$, then $(\omega)^z_{\psi,G}$ holds for $z = (z_1, z_2)$ and $\psi = \psi_1 \times \psi_2$.

In this way, we can reduce the study of $(\omega)^z_{\psi,G}$ to the cases where W is irreducible as a Coxeter group. In particular, we can reduce the cases where W is a product of symmetric groups and $\omega \in \{\triangle, +\}$ to Conjecture A.

To answer question (I), we prove results that directly generalize the three parts of [Hai93, Proposition 4.1].

In [Tri21], we proved a result essentially stating that $\tau_{z,\mathsf{v}}$ intertwines parabolic inclusions $W' \subseteq W$ with induction of characters from W' to W. In Section 5, we prove that the characters $spr_{\psi,G}$ enjoy a similar compatibility with induction (Theorem 5.3), refining an observation of Alvis–Lusztig: This may be of independent interest. From combining these "induction theorems", we obtain the result below: in fact, a more precise form generalizing part (2) of [Hai93, Proposition 4.1].

Proposition 1.9. *If z lies in a parabolic subgroup $W' \subseteq W$, corresponding to a Levi subgroup $G' \subseteq G$, then the $\alpha^z_{\psi,G}$ for $\psi \in \mathrm{Irr}(W)$ are nonnegative integer sums of the $\alpha^z_{\psi',G'}$ for $\psi' \in \mathrm{Irr}(W')$. In particular, if $\omega \in \{\triangle, \pm, +\}$ and $(\omega)^z_{\psi',G'}$ holds for all $\psi' \in \mathrm{Irr}(W')$, then $(\omega)^z_{\psi,G}$ holds for all $\psi \in \mathrm{Irr}(W)$.*

We can also give a sufficient condition for the $\alpha^z_{\psi,G}$ to vanish. Recall that the Kazhdan–Lusztig basis defines a partition of W into subsets $\mathcal{F}$ called *two-sided cells*, arranged in a partial order. Each $\mathcal{F}$ gives rise to a module over the Hecke algebra, and each simple Hecke module occurs in exactly one such module, producing a map $\chi \mapsto \mathcal{F}(\chi)$ from irreducible characters of W to two-sided cells. When $W = S_n$, this map is a bijection, and the partial order on two-sided cells recovers the reverse dominance order on partitions of n. So the following result, explained in Section 6, generalizes part (3) of [Hai93, Proposition 4.1].

Proposition 1.10. *If $\psi \in \mathrm{Irr}(W)$ and $z \in \mathcal{F}$ for some two-sided cell $\mathcal{F} < \mathcal{F}(\psi)$, then $\alpha^z_{\psi,G}(\mathsf{v}) = 0$.*

Let $w_\circ$ be the longest element of W. The following result generalizes part (1) of [Hai93, Proposition 4.1].

Corollary 1.11. *For the trivial character 1_W, we have*

$$\alpha^{w_\circ}_{1_W,G}(\mathsf{v}) = \sum_{w \in W} \mathsf{v}^{2\ell(w)-\ell(w_\circ)}.$$

If ψ is nontrivial, then $\alpha^{w_\circ}_{\psi,G}(\mathsf{v}) = 0$.

It seems tricky to formulate other uniform results about question (I). For example, we do not know how to generalize [Hai93, Proposition 4.2], which concerns the case where z is a Coxeter element: For each Coxeter element z in type D_4, it turns out that $(+)^z_{\psi,G}$ fails when ψ is the character labeled 1.3 in CHEVIE.

As another example: At an earlier stage of this work, we had conjectured that if $W = W(B_n) = W(C_n)$ for some n and z is rationally smooth, then $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ hold for all ψ. This is true up through rank 6, but both properties fail in both type B_7 and type C_7 for many rationally smooth z. In the cases where $(\pm)^z_{\psi,G}$ fails, $\ell(z) \geq 18$.

1.4. **A New Conjecture.** Question (II) led us to Conjecture B below. It suggests that some compatibility between inflation maps and the Springer correspondence allows the positivity and unimodality in Haiman's Conjecture A to be lifted to other Weyl groups.

Observe that there is a unique quotient map $W(G_2) \to W(A_2) = S_3$ sending simple reflections to simple reflections. We call it the *G_2-to-A_2 quotient*.

Conjecture B. *Suppose that $\psi \in \mathrm{Irr}(W)$ is inflated from a quotient $\varphi \colon W \to \Omega$. If Ω is isomorphic to a product of symmetric groups, then $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ hold for all $z \in W$. Moreover, if φ does not descend to the G_2-to-A_2 quotient, then $(+)^z_{\psi,G}$ holds for all z.*

Example 1.12. The trivial and sign characters of W are inflated from $S_2 \simeq \{\pm 1\}$ along the sign character itself. So Conjecture B predicts that if ψ is either of these characters, then $(\triangle)^z_{\psi,G}$ and $(+)^z_{\psi,G}$ hold for all z.

For the trivial character, we do not know a proof of these statements. Compare to [Hai93, Conjecture 4.1].

For the sign character ε, we claim that $\alpha^e_{\varepsilon,G} = 1$ and $\alpha^z_{\varepsilon,G} = 0$ for all $z \neq e$. The latter is part (1) of Corollary 6.4. As for the former, Theorem 1.1 and the discussion in §2.2 show that $\tau_{e,\mathsf{v}}$ is the character of the regular representation of W, which coincides with $spr_{\varepsilon,G}$. Thus, $\tau_{e,\mathsf{v}} = spr_{\varepsilon,G}$, giving $\alpha^e_{\varepsilon,G} = 1$.

Maxwell showed [Max98] that if W is an irreducible finite Coxeter group and $\varphi\colon W \to \Omega$ is a quotient map, then either or both of the following must hold:

- The longest element $w_\circ \in W$ is central, and φ is reduction modulo $w_\circ$.
- Ω is another Coxeter group, and φ is a *graph homomorphism*[3]: that is, a quotient map that sends each simple reflection of W either to a simple reflection of Ω or to the identity of Ω.

Example 1.13. Take $W = W(B_n) = W(C_n)$. Label the simple reflections of W by $s_1, s_2, \ldots, s_n$, where $(s_1 s_2)^4 = e$ and $(s_i s_{i+1})^3 = e$ for $i \geq 2$. Then reduction modulo ts_1ts_1 is a graph homomorphism $\varphi_{1212}\colon W \to W(A_1 \times A_{n-1}) = S_2 \times S_n$.

Following [Car93], we index the irreducible characters of W by bipartitions of n: ordered pairs of partitions (ξ, η) such that $n = |\xi| + |\eta|$. Writing $\psi_{\xi,\eta}$ for the associated character, one checks that the following conditions are equivalent:

- $\psi_{\xi,\eta}$ is inflated from $S_2 \times S_n$ along φ_{1212}.
- $\psi_{\xi,\eta}$ is inflated from a product of symmetric groups.
- One of ξ or η is empty.

Examining the third condition is how we discovered Conjecture B.

For instance, when $n = 4$, there are 20 irreducible characters total. Of these, 10 are inflated along φ_{1212}. Of the remainder, there are 3 where $(+)^z_{\psi,G}$ fails in type B_4 for some z, and 4 where $(+)^z_{\psi,G}$ fails in type C_4 for some z.

By Remark 1.8, it is enough to check Conjecture B in the cases where W is irreducible. In Section 7, we summarize the verification of Conjecture B for the following cases, using the GAP3 package CHEVIE [GHL+96]:

(1) W irreducible of rank ≤ 5.
(2) W irreducible of rank ≤ 6, where we restrict z to be rationally smooth.

As explained there, the data and scripts used in the verification are available at a webpage that also contains partial data in types B_7, C_7, D_7.

There are other predictions that one could draw from the evidence in Section 7, but perhaps none as striking as Conjecture B. For example: Given $\omega \in \{\triangle, \pm, +\}$, let $\mathrm{Irr}(W)_{G,\omega}$ be the set of $\psi \in \mathrm{Irr}(W)$ where $(\omega)^z_{\psi,G}$ holds for all $z \in W$. In our

[3]So named because the graph homomorphisms from (W, S) onto itself are the automorphisms induced by symmetries of its Dynkin diagram.

examples, the sets $\mathrm{Irr}(W)_{G,\triangle}$ and $\mathrm{Irr}(W)_{G,\pm}$ are often equal, but not always; one may contain the other as a strict subset. We have not found any cases where they are incomparable. Moreover, they are noticeably larger than $\mathrm{Irr}(W)_{G,+}$. It is necessarily true that $\mathrm{Irr}(W)_{G,+} \subseteq \mathrm{Irr}(W)_{G,\pm}$, but it is surprising to find that we always have $\mathrm{Irr}(W)_{G,+} \subseteq \mathrm{Irr}(W)_{G,\triangle}$ as well. We expect that if ψ is fixed, and $(\triangle)^z_{\psi,G}$ or $(\pm)^z_{\psi,G}$ fails for some $z \in W$, then there is always some $z' \in W$ where the nonzero coefficients of $\alpha^{z'}_{\psi,G}$ are all *negative*.

1.5. **Further Directions.** There are at least two.

1.5.1. *From Finite to Affine.* Affinization is the principle that structures for a complex reductive group G often have analogues for its loop group LG or the associated affine Kac–Moody group.

The affine analogue of a Lusztig variety was first studied in [Lus11] and [KV12]. The term *affine Lusztig variety* was introduced by X. He [He25], although other names have been used, such as *generalized affine Springer fiber*, *multiplicative affine Springer fiber*, and *Kottwitz–Viehmann variety*. Just as Lusztig varieties originally appear in the theory of character sheaves, so affine Lusztig varieties are expected to appear in a putative theory of character sheaves on loop groups.

Since affine Lusztig varieties for regular semisimple elements of the loop group LG are still finite-dimensional, we can use them to define an analogue of $\tau_{z,\vee}$ geometrically. It should take the form of a class function on an extended affine Weyl group. However, it is less clear what should replace the functions $spr_{\psi,G}$, since affine Springer fibers for unipotent elements of LG are infinite-dimensional. It seems plausible that one should use the ungraded characters of affine Springer representations arising from *topologically* unipotent elements of LG. Here, the analogues of the groups A_u and their characters κ play a major role in the arithmetic of the affine Springer fibers: more precisely, in the theory of endoscopy in the Langlands program [Wan26].

Question 1.14. Is there an analogue of Conjecture B, relating the intersection cohomology of affine Lusztig varieties to ungraded affine Springer characters?

1.5.2. *From Weyl Groups to Coxeter Groups.* In [Lus93, Lus94], Lusztig and Malle showed that certain postulates uniquely determine an analogue of the exotic Fourier transform from [Lus84] for any finite Coxeter group. (More accurately, Malle's appendix to [Lus94] handles type H_4.) Consequently, $\tau_{z,\vee}$ can be generalized from Weyl groups to all finite groups.

At the same time, in [AA08], Achar–Aubert showed that certain axioms determine what they call *preferred* analogues of the map $\psi \mapsto spr_{\psi,G}$ for dihedral groups. In [Ach09], Achar reports on attempts to extend the characters $spr_{\psi,G}$ even further, to complex reflection groups.

Question 1.15. Can we uniformly generalize Conjecture B to finite Coxeter groups?

1.6. **Acknowledgments.** I thank Patrick Brosnan, Jaehyun Hong, Donggun Lee, and Eunjeong Lee for a discussion held shortly after I had sent them an early draft of this paper. Independently, they too had observed Theorem 1.1, and had used GAP3 and Sage to compute the Laurent polynomials $\alpha^z_{\psi,G}(\mathsf{v})$ in low rank. Moreover, in the draft I had sent, I had only claimed Theorem 1.1 for g in a dense open locus of G^{rs}. I am grateful to them for pointing out that Abreu–Nigro's work allows us to extend that locus to G^{rs} itself.

I thank Roman Bezrukavnikov for a discussion of induction theorems for Springer-fiber cohomologies that helped me to arrive at Theorem 5.3.

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2. Setup

This section reviews the geometry needed in Sections 3 and 4.

Note that, while Section 1 discussed algebraic varieties over the complex numbers, Lusztig's papers on character sheaves start from a general algebraically closed field $\mathbf{k}$. We do the same, and specialize $\mathbf{k}$ to the field of complex numbers or to the algebraic closure of a finite field only when needed. We fix a prime ℓ invertible in $\mathbf{k}$, and work with $\bar{\mathbf{Q}}_\ell$-complexes in the étale topology on $\mathbf{k}$-varieties.

Let G be a connected reductive algebraic group over $\mathbf{k}$ with Weyl group W. The characteristic of $\mathbf{k}$ is either zero or a good prime for G [Car93, 28].

2.1. Unipotent Character Sheaves.

Let $\mathcal{B}$ be the flag variety of G. Let

$$act\colon \mathcal{B} \times G \to \mathcal{B} \times \mathcal{B}$$

be defined by $act(B, g) = (B, g^{-1}Bg)$. For all $z \in W$, let $O_z \subseteq \mathcal{B} \times \mathcal{B}$ be the G-orbit of pairs in relative position z. Sheaves on these orbits form a geometrization of the Hecke algebra H_W, while pullback-pushforward through the correspondence

$$\mathcal{B} \times \mathcal{B} \xleftarrow{act} \mathcal{B} \times G \xrightarrow{pr} G$$

is a geometrization of its cocenter map.

Our notations will mostly follow Lusztig's notations in [Lus85b]. Let $Y_z \subseteq \mathcal{B} \times G$ be defined by the pullback square:

$$\begin{tikzcd} O_z \arrow[d] & Y_z \arrow[l] \arrow[d] \\ \mathcal{B} \times \mathcal{B} & \mathcal{B} \times G \arrow[l, "act"'] \end{tikzcd}$$

Let $\bar{Y}_z$ be the Zariski closure of Y_z in $\mathcal{B} \times G$. Thus $\bar{Y}_z = \bigcup_{w \leq v} Y_w$, where $\leq$ is the Bruhat order on W corresponding to the closure order on the orbits O_w.

Let $j\colon Y_z \to \bar{Y}_z$ be the inclusion map. Let $\pi_z\colon Y_z \to G$ and $\bar{\pi}_z\colon \bar{Y}_z \to G$ be the projection maps that factor through pr.

Let $j_{!*}$, denote intermediate extension of shifted perverse sheaves along j [Ach21, §5.1]. Let $J_z = j_{!*}\underline{\bar{\mathbf{Q}}_\ell}_{Y_z}$, the (unshifted) intersection cohomology complex of $\bar{Y}_z$. Let $\bar{K}_z = \bar{\pi}_{z,!}J_z = \bar{\pi}_{z,*}J_z$. By the decomposition theorem [Ach21, §3.9], there is a noncanonical isomorphism

$$\bar{K}_z \simeq \bigoplus_i {}^p\mathcal{H}^i(\bar{K}_z)[-i],$$

where ${}^p\mathcal{H}^i$ denotes the ith cohomology sheaf with respect to the perverse t-structure, and each perverse sheaf ${}^p\mathcal{H}^i(\bar{K}_z)$ is semisimple. The simple summands of these perverse sheaves, as we run over z and i, are called *unipotent character sheaves*.

2.2. **The Grothendieck Alteration.** The map $\pi_e = \bar{\pi}_e$ is called the *Grothendieck alteration* of G. Below, we review facts about π_e that can be proved in an analogous way to their counterparts for the Lie algebra of G [Ach21, §8.2].

First, the pullback of π_e to the regular semisimple locus G^{rs} is a Galois étale cover with Galois group W. In particular, if $\mathbf{k} = \mathbf{C}$, then W is a quotient of the topological fundamental group of G^{rs}.

Next, π_e is a small map. So $\bar{K}_e$ is the intermediate extension of its restriction $\bar{K}_e|_{G^{rs}}$, and the Galois action of W on $\bar{K}_e|_{G^{rs}}$ induces one on $\bar{K}_e$ by functoriality. We obtain a W-equivariant decomposition

$$\bar{K}_e[\dim G] \simeq \bigoplus_{\chi \in \mathrm{Irr}(W)} V_\chi \otimes \mathcal{A}_\chi, \tag{2.1}$$

where the $\mathcal{A}_\chi$ are simple perverse sheaves and $V_\chi := \mathrm{Hom}(\mathcal{A}_\chi, \bar{K}_e[\dim G])$ affords a representation of W with character χ.

A unipotent character sheaf that is not isomorphic to $\mathcal{A}_\chi$ for any χ is called *cuspidal* [Lus85a].

Theorem 2.1 (Lusztig)**.** *Suppose that $\mathbf{k}$ is either of characteristic zero or the algebraic closure of a finite field. Then for any $z \in W$ and $\chi \in \mathrm{Irr}(W)$, we have*

$$\sum_i (-\mathsf{v})^i (\mathcal{A}_\chi : {}^p\mathcal{H}^i(\bar{K}_z)) = (-1)^{\dim G} \mathsf{v}^{\dim G + \ell(z)} \sum_{\psi \in \mathrm{Irr}(W)} \{\chi, \psi\} \psi_{\mathsf{v}}(c_z),$$

where ℓ is Bruhat length, as in §1.2, and $\{-,-\}$ is the exotic Fourier transform, as in Appendix A.

Proof. When G is defined over a finite field and $\mathbf{k}$ is its algebraic closure, this follows from Lusztig's [Lus85b, Corollary 14.11] and [Lus86, Theorem 23.1(c)]. See also the explanation of [WW11, 413].

When $\mathbf{k}$ is of characteristic zero, we deduce the result by a standard "spreading out" argument, following [Spr78, §2.7] and the references to SGA 4 there. In brief, we spread out G and the relevant $\bar{\mathbf{Q}}_\ell$-complexes to a model over a finitely generated ring $R \subseteq \mathbf{k}$ in which ℓ is invertible, using the specialization theorem for étale cohomology. After shrinking R, we may assume that the complexes are locally constant over $\mathrm{Spec}(R)$. At some closed point $\mathfrak{p} \in \mathrm{Spec}(R)$ and geometric point $\bar{\mathfrak{p}}$ over $\mathfrak{p}$, the multiplicity formula holds at $\bar{\mathfrak{p}}$. Hence, by the comparison theorem for étale cohomology, the same formula holds at the geometric point of $\mathrm{Spec}(R)$ corresponding to the embedding of R into $\mathbf{k}$. $\square$

2.3. **The Springer Resolution.** Let G^{uni} be the variety of unipotent elements of G. Recall that $\dim G^{uni} = 2N$, where N is the number of positive roots of G. Let $\pi_z^{uni} \colon Y_z^{uni} \to G^{uni}$ and $\bar{\pi}_z^{uni} \colon \bar{Y}_z^{uni} \to G^{uni}$ be defined by the pullback squares:

$$\begin{tikzcd} Y_z^{uni} \arrow[r] \arrow[d, "\pi_z^{uni}"'] & Y_z \arrow[d, "\pi_z"] \\ G^{uni} \arrow[r] & G \end{tikzcd} \qquad \begin{tikzcd} \bar{Y}_z^{uni} \arrow[r] \arrow[d, "\bar{\pi}_z^{uni}"'] & \bar{Y}_z \arrow[d, "\bar{\pi}_z"] \\ G^{uni} \arrow[r] & G \end{tikzcd}$$

As mentioned in §1.2, $\pi_e^{uni} = \bar{\pi}_e^{uni}$ is called the *Springer resolution* of G^{uni}.

The *Springer sheaf* is $\mathcal{S} := \pi_{e,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{Y_e^{uni}}[2N] = \pi_{e,*}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{Y_e^{uni}}[2N]$. Since Y_e is smooth, being smooth over $\mathcal{B}$, we have $\mathcal{S}[-2N] = \pi_{e,!} \underline{\bar{\mathbf{Q}}_\ell}_{Y_e}|_{G^{uni}} = \bar{K}_e|_{G^{uni}}$. Thus (2.1) restricts to a decomposition

$$\mathcal{S} \simeq \bigoplus_{\chi \in \mathrm{Irr}(W)} V_\chi \otimes \mathcal{S}_\chi, \tag{2.2}$$

where $\mathcal{S}_\chi = \mathcal{A}_\chi[2N - \dim G]|_{G^{uni}}$. In particular, the W-action on $\bar{K}_e$ restricts to an action on $\mathcal{S}$, for which we have W-equivariant isomorphisms $V_\chi \simeq \mathrm{Hom}(\mathcal{S}_\chi, \mathcal{S})$.

In Section 4, we actually need G-equivariant versions of the Springer sheaf and its summands. Let G act on itself by (right) conjugation. Then Y_z, $\bar{Y}_z$ are stable under the G-action on $\mathcal{B} \times G$. Since G^{uni} is stable under conjugation, Y_z^{uni}, $\bar{Y}_z^{uni}$ are also stable under the G-action on $\mathcal{B} \times G$. So the maps π_z^{uni}, $\bar{\pi}_z^{uni}$ are equivariant. In particular, $\mathcal{S}$ and its summands $\mathcal{S}_\chi$ can be upgraded to objects of the G-equivariant derived category of $\bar{\mathbf{Q}}_\ell$-complexes on G^{uni} [Ach21, §6.4]. We denote the equivariant version of $\mathcal{S}$ by $\mathcal{S}_G$.

In any G-equivariant derived category of $\bar{\mathbf{Q}}_\ell$-complexes, we write $\mathrm{Hom}^n(K, K') = \mathrm{Hom}(K, K'[n])$. The following is an analogue of [Lus88, Corollary 6.4], with G^{uni} in place of the nilpotent cone.

Theorem 2.2 (Lusztig)**.** (1) $\mathrm{Hom}^i(\mathcal{S}_G, \mathcal{S}_G)$ *vanishes in odd degrees* i.

(2) *There is an isomorphism of graded* $\bar{\mathbf{Q}}_\ell$*-algebras*

$$\mathrm{Hom}^{2*}(\mathcal{S}_G, \mathcal{S}_G) \simeq \mathrm{Sym}^*(\bar{\mathbf{Q}}_\ell \otimes \mathbb{X}) \rtimes \bar{\mathbf{Q}}_\ell W, \tag{2.3}$$

where $\mathbb{X}$ *is the character lattice in the root datum of* G*, as in §1.2, and* $\bar{\mathbf{Q}}_\ell W$ *is placed in degree* 0*. The degree-*0 *isomorphism* $\mathrm{Hom}^0(\mathcal{S}_G, \mathcal{S}_G) \simeq \bar{\mathbf{Q}}_\ell W$ *recovers the* W*-action on* $\mathcal{S}_G$ *arising from* (2.2)*.*

As in [Tri21], we may identify $\mathrm{Hom}^*(\mathcal{S}_G, \mathcal{S}_G)$ with the convolution algebra formed by the equivariant Borel–Moore homology of the Steinberg variety from (1.2).

3. The Regular Semisimple Locus

The goal of this section is to prove Theorem 1.1. We keep the hypotheses and notation of Section 2, but now, take $\mathbf{k} = \mathbf{C}$.

Recall that IH^* denotes étale intersection cohomology with $\bar{\mathbf{Q}}_\ell$-coefficients, and that $\bar{Y}_{z,g}$ is the fiber of $\bar{Y}_z$ above $g \in G(\mathbf{C})$. For any $\bar{\mathbf{Q}}_\ell$-complex K on G and $g \in G(\mathbf{C})$, we write $\mathcal{H}^*_g(K)$ for the stalk at g of the cohomology sheaf $\mathcal{H}^*(K)$.

Lemma 3.1. *For any $z \in W$ and $g \in G^{rs}(\mathbf{C})$, we have a W-equivariant isomorphism*

$$\mathrm{IH}^*(\bar{Y}_{z,g}) \simeq \mathcal{H}^*_g(\bar{K}_z).$$

Proof. Let $i_g \colon \bar{Y}_{z,g} \to \bar{Y}_z$ be the inclusion map. Let $j_g \colon Y_{z,g} \to \bar{Y}_{z,g}$ be the inclusion map, parallel to our earlier notation $j \colon Y_z \to \bar{Y}_z$. Then

$$\mathrm{IH}^*(\bar{Y}_{z,g}) = \mathrm{H}^*(\bar{Y}_{z,g}, j_{g,!*} i_g^* \underline{\mathbf{C}}_{Y_z}),$$
$$\mathcal{H}^*_g(\bar{K}_z) = \mathrm{H}^*(\bar{Y}_{z,g}, i_g^* j_{!*} \underline{\mathbf{C}}_{Y_z}).$$

The usual base change theorem gives a $\pi_1(G^{rs}, g)$-equivariant isomorphism of (derived) functors $i_g^* j_! \xrightarrow{\sim} j_{g,!} i_g^*$. So it suffices to show that the composition $i_g^* j_! \xrightarrow{\sim} j_{g,!} i_g^* \to j_{g,!*} i_g^*$ factors through an isomorphism $i_g^* j_{!*} \xrightarrow{\sim} j_{g,!*} i_g^*$.

This holds on some Zariski dense open subset of G^{rs} by the "generic base change" theorem of [Del77, Théorème 1.9].[4] But by [AN25, Proposition 8.2], $\bar{Y}_z(\mathbf{C}) \to G(\mathbf{C})$ restricts to a topological fibration over $G^{rs}(\mathbf{C})$.[5] Therefore, the desired isomorphism holds for all $g \in G^{rs}(\mathbf{C})$. □

The preceding lemma reduces the study of $\mathrm{IH}^*(\bar{Y}_{z,g})$ to the study of $\mathcal{H}^*_g(\mathcal{A})$ for unipotent character sheaves $\mathcal{A}$.

Lemma 3.2. *For any $\chi \in \mathrm{Irr}(W)$ and $g \in G^{rs}(\mathbf{C})$, the graded vector space $\mathcal{H}^*_g(\mathcal{A}_\chi)$ is concentrated in degree $-\dim G$. The monodromy action of $\pi_1(G^{rs}, g)$ on it defines a representation of W with character χ.*

Proof. Let $Y_e^{rs} \to G^{rs}$ be the pullback of $Y_e \to G$. Since $Y_e^{rs} \to G^{rs}$ is a W-torsor, and W is finite, we know that $\mathcal{H}^*_g(\bar{K}_e)$ is concentrated in degree 0. Since $\mathcal{A}_\chi$ is a summand of $\bar{K}_e[\dim G]$, we deduce the first claim.

By construction, the monodromy action on the W-torsor $\mathcal{H}^0_g(\bar{K}_e)$ factors through the regular representation of W. The resulting W-action commutes with the one arising from the Galois action on $\bar{K}_e$. So, from comparing (2.1) to the bi-equivariant decomposition $\mathbf{C}W \simeq \bigoplus_\chi V_\chi \otimes V_{\chi^\vee}$, we deduce that the monodromy action on $\mathcal{H}^*_g(\mathcal{A}_\chi)$ has character $\chi^\vee$, where $(-)^\vee$ denotes duals. But irreducible characters of Weyl groups are real-valued, so they are self-dual. □

Lemma 3.3 (Lusztig)**.** *If $\mathcal{A}$ is a cuspidal unipotent character sheaf on G, then $\mathcal{H}^*_g(\mathcal{A})$ vanishes for all $g \in G^{rs}(\mathbf{C})$.*

[4] See also https://mathoverflow.net/a/123787.

[5] I thank Patrick Brosnan, Jaehyun Hong, Donggun Lee, and Eunjeong Lee for pointing out this reference.

Proof. Let $Z_G \subseteq G$ be the center and $Z_G^\circ \subseteq Z_G$ its neutral component.

By [Lus85a, (7.1.2)], $\mathcal{A} = \mathrm{IC}(\bar{\Sigma}, \mathcal{E})[d]$ for some $(G \times Z_G^\circ)$-orbit $\Sigma \subseteq G$, some simple $(G \times Z_G^\circ)$-equivariant local system $\mathcal{E}$ on Σ, and some integer d. Moreover, by [Lus86, Theorem 23.1(a)], $\mathcal{A}$ is *clean*: Its restriction to $\bar{\Sigma} - \Sigma$ is zero. (Strictly speaking, *loc. cit.* is for G over the algebraic closure of a finite field. We deduce the result over $\mathbf{C}$ by spreading out, as in the proof of Theorem 2.1. Note that *loc. cit.* relies on definitions in [Lus85b, (13.9.2)] and [Lus85a, (7.7)].) It remains to show that Σ cannot contain g.

If G is a torus, then the only unipotent character sheaf is the constant sheaf, so G cannot be a torus. So G_g/Z_G° must be nontrivial, where $G_g \subseteq G$ is the centralizer of g. But if $g \in \Sigma$, then [Lus85a, (7.1.2)] shows that G_g/Z_G° must be unipotent, whereas G_g is a torus. □

Proof of Theorem 1.1. Fix $g \in G^{rs}(\mathbf{C})$ and $w \in W$. We are claiming that

$$\mathsf{v}^{\ell(z)}\tau_{z,\mathsf{v}}(w) = \sum_i \mathsf{v}^{2i}\,\mathrm{tr}(w \mid \mathrm{IH}^{2i}(\bar{Y}_{z,g})), \tag{3.1}$$

where the W-action on $\mathrm{IH}^*(\bar{Y}_{z,g})$ arises from the monodromy action of $\pi_1(G^{rs}, g)$. In the ring of polynomials in v over the representation ring of W with $\bar{\mathbf{Q}}_\ell$-coefficients, we have

$$\begin{aligned}
&\sum_j (-\mathsf{v})^j\,\mathrm{IH}^j(\bar{Y}_{z,g}) \\
&= \sum_j (-\mathsf{v})^j\,\mathcal{H}_g^j(\bar{K}_z) && \text{by Lemma 3.1} \\
&= \sum_{i,j} (-\mathsf{v})^j\,\mathcal{H}_g^{j-i}({}^p\mathcal{H}^i(\bar{K}_z)) && \text{by the decomposition theorem} \\
&= \sum_k (-\mathsf{v})^k\,\mathcal{H}_g^k\left(\sum_i (-\mathsf{v})^i\,{}^p\mathcal{H}^i(\bar{K}_z)\right) \\
&= \sum_k (-\mathsf{v})^k\,\mathcal{H}_g^k\left(\sum_{i,\chi} (-\mathsf{v})^i(\mathcal{A}_\chi : {}^p\mathcal{H}^i(\bar{K}_z))\mathcal{A}_\chi\right) && \text{by Lemma 3.3} \\
&= \mathsf{v}^{\ell(z)} \sum_{k,\chi,\psi} (-\mathsf{v})^{k+\dim G}\{\chi,\psi\}\psi_\mathsf{v}(c_z)\,\mathcal{H}_g^k(\mathcal{A}_\chi) && \text{by Theorem 2.1.}
\end{aligned}$$

By Lemma 3.2, $\mathcal{H}_g^*(\mathcal{A}_\chi)$ is concentrated in degree $-\dim G$, and the monodromy action on it factors through a representation of W with character χ. Combining this with the preceding chain of equalities,

$$\mathsf{v}^{\ell(z)} \sum_{\chi,\psi} \{\chi,\psi\}\psi_\mathsf{v}(c_z)\chi(w) = \sum_j (-\mathsf{v})^j\ \mathrm{tr}(w \mid \mathrm{IH}^j(\bar{Y}_{z,g})).$$

The left-hand side is $\mathsf{v}^{\ell(z)}\tau_{z,\mathsf{v}}(w)$.

Finally, Theorem 1.6(1) and display (9.3) in [AN25] together show that when g is regular semisimple, $\mathrm{IH}^*(\bar{Y}_{z,g})$ vanishes in odd degrees and $\dim \bar{Y}_{z,g} = \dim Y_{z,g} = \ell(z)$, as mentioned in §1.2. So the right-hand side of the last display simplifies to the right-hand side of (3.1). ☐

Remark 3.4. Fix a parabolic subgroup $W' \subseteq W$. Abreu–Nigro's [AN25, Theorem 1.5] asserts that for all $z \in W$ and $g \in G^{rs}(\mathbf{C})$, we have

$$\mathsf{v}^{\ell(z)} \sum_{\psi} \psi_{\mathsf{v}}(c_z) \dim V_{\psi}^{W'} = \sum_i \mathsf{v}^{2i} \dim \mathrm{IH}^{2i}(\bar{Y}_{z,g})^{W'}.$$

By comparing with Theorem 1.1, and using the fact that the character table of the Hecke algebra specializes to that of W, which is invertible, we deduce the following curious property of the exotic Fourier transform:

$$\sum_{\chi} \{\chi, \psi\} \dim V_{\chi}^{W'} = \dim V_{\psi}^{W'} \quad \text{for all } \psi \in \mathrm{Irr}(W).$$

Compare to the *tower equivalence* property of $\{-,-\}$ discussed in [Mic23].

4. The Unipotent Locus

The goal of this section is to prove Theorem 1.3 as well as its generalization to arbitrary $z \in W$. We keep the hypotheses and notation of Section 2.

We write K_G for the equivariant version of any G-stable $\bar{\mathbf{Q}}_\ell$-complex K on a scheme of finite type with a G-action. Recall that $\mathrm{H}^*_{c,G}$ denotes G-equivariant compactly-supported étale cohomology with $\bar{\mathbf{Q}}_\ell$-coefficients, and that we set

$$\begin{aligned}\bar{\mathcal{Z}}_z &= Y_e^{uni} \times_G \bar{Y}_z \\ &\simeq Y_e^{uni} \times_{G^{uni}} \bar{Y}_z^{uni}.\end{aligned}$$

Recall from §2.3 that we write $2N = \dim G^{uni} = \dim Y_e^{uni}$.

4.1. Rational Smoothness. An element $z \in W$ is *rationally smooth* when the intersection cohomology complex of $\bar{O}_z$, the Zariski closure of O_z in $\mathcal{B} \times \mathcal{B}$, is a constant sheaf. In this case, the intersection cohomology complex of $\bar{Y}_z$ is also constant: $J_z = \underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z}$. This, in turn, implies that $\bar{K}_z = \bar{\pi}_{z,!}\underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z} = \bar{\pi}_{z,*}\underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z}$.

Lemma 4.1. *For any $z \in W$ and integer i, we have*

$$\mathrm{H}^i_{c,G}(\bar{\mathcal{Z}}_z)^\vee \simeq \mathrm{Hom}^{2N-i}(\bar{\pi}^{uni}_{z,!}\underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z^{uni},G}, \mathcal{S}_G),$$

realizing the graded action of $\mathrm{Hom}^*(\mathcal{S}_G, \mathcal{S}_G)$ *on* $\mathrm{H}^i_{c,G}(\bar{\mathcal{Z}}_z)$ *via (derived) composition. In particular, if z is rationally smooth, then*

$$\mathrm{H}^i_{c,G}(\bar{\mathcal{Z}}_z)^\vee \simeq \mathrm{Hom}^{2N-i}(\bar{K}_{z,G}|_{G^{uni}}, \mathcal{S}).$$

Proof. Let $\bar{\Pi}_z^{uni} \colon \bar{\mathcal{Z}}_z \to Y_e^{uni}$ be the pullback of $\bar{\pi}_z^{uni} \colon \bar{Y}_z^{uni} \to G^{uni}$. Let $\omega_{Y_e^{uni},G}$ be the G-equivariant dualizing complex of Y_e^{uni}. We compute

$$\begin{aligned}
\mathrm{H}_c^i(\bar{\mathcal{Z}}_z)^\vee &= \mathrm{H}_c^i(Y_e^{uni}, \bar{\Pi}_{z,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{\bar{\mathcal{Z}}_z,G})^\vee \\
&\simeq \mathrm{Hom}^{-i}(\bar{\Pi}_{z,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{\bar{\mathcal{Z}}_z,G}, \omega_{Y_e^{uni},G}) && \text{by Verdier} \\
&\simeq \mathrm{Hom}^{4N-i}(\bar{\Pi}_{z,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{\bar{\mathcal{Z}}_z,G}, \underline{\bar{\mathbf{Q}}_\ell}_{Y_e^{uni},G}) && \text{by smoothness of } Y_e^{uni} \\
&\simeq \mathrm{Hom}^{4N-i}(\pi_e^* \bar{\pi}_{z,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z^{uni},G}, \underline{\bar{\mathbf{Q}}_\ell}_{Y_e^{uni},G}) && \text{by base change} \\
&\simeq \mathrm{Hom}^{2N-i}(\bar{\pi}_{z,!}^{uni} \underline{\bar{\mathbf{Q}}_\ell}_{\bar{Y}_z^{uni},G}, \mathcal{S}_G),
\end{aligned}$$

as claimed. □

The preceding lemma reduces the study of $\mathrm{H}_{c,G}^*(\bar{\mathcal{Z}}_z)$ to that of $\mathrm{Hom}^*(\mathcal{A}_G|_{G^{uni}}, \mathcal{S}_G)$ for unipotent character sheaves $\mathcal{A}$. Note that since $\bar{\pi}_z$ is G-equivariant, $\bar{K}_z$ and its shifted perverse summands are indeed G-stable, so the equivariant $\bar{\mathbf{Q}}_\ell$-complexes $\mathcal{A}_G$ are indeed well-defined.

Recall the complexes $\mathcal{S}_\chi = \mathcal{A}_\chi[2N - \dim G]|_{G^{uni}}$ in (2.2). Since the left W^{op}-action on the right-hand side of (2.3) corresponds to the left W^{op}-action on the left-hand side arising from (2.2), Theorem 2.2 implies:

Lemma 4.2 (Lusztig). (1) $\mathrm{Hom}^i(\mathcal{S}_{\chi,G}, \mathcal{S}_G)$ *vanishes in odd degrees* i.

(2) *For all* j*, there is a* W*-equivariant isomorphism*

$$\mathrm{Hom}^{2j}(\mathcal{S}_{\chi,G}, \mathcal{S}_G) \simeq \mathrm{Sym}^j(\bar{\mathbf{Q}}_\ell \otimes \mathbb{X}) \otimes V_\chi,$$

where the W*-action on the left-hand side arises from* $\mathrm{Hom}^0(\mathcal{S}_G, \mathcal{S}_G) \simeq \bar{\mathbf{Q}}_\ell W$.

Now suppose that the unipotent character sheaf $\mathcal{A}$ is cuspidal. By reducing to the case where G is almost simple and adjoint, and using the classification of cuspidal character sheaves on such groups in [Lus86], one can check that $\mathcal{A}|_{G^{uni}}$ is always either zero or a shift of a perverse sheaf on G^{uni} that is *cuspidal* in the sense of [RR16, Definition 2.3]: For details, see [Aub98, §2]. Thus, [RR16, Theorem 3.5], or rather its analogue with G^{uni} in place of the nilpotent cone, implies:

Lemma 4.3 (Lusztig, Rider–Russell). *If* $\mathcal{A}$ *is a cuspidal unipotent character sheaf on* G*, then* $\mathrm{Hom}^*(\mathcal{A}_G|_{G^{uni}}, \mathcal{S}_G)$ *vanishes.*

Proof of Theorem 1.3. Suppose that z is rationally smooth, and fix $w \in W$. We are claiming that

$$\sum_j (-\mathsf{v})^j \,\mathrm{tr}(w \mid \mathrm{H}_{c,G}^j(\bar{\mathcal{Z}}_z)) = (-1)^{\ell(w)} \mathsf{v}^{2\dim G + \ell(z)} \mathbb{S}_{\mathsf{v}^2}(w) \tau_{z,\mathsf{v}}(w),$$

where the W-action on $\mathrm{H}_{c,G}^*(\bar{\mathcal{Z}}_z)$ arises from the action of $\mathrm{Hom}^0(\mathcal{S}_G, \mathcal{S}_G)$. Since characters of Weyl groups are self-dual, the trace of w on $\mathrm{H}_{c,G}^j(\bar{\mathcal{Z}}_z)$ is also its trace on $\mathrm{H}_{c,G}^j(\bar{\mathcal{Z}}_z)^\vee$. In the ring of formal power series in v over the representation ring of

W with $\bar{\mathbf{Q}}_\ell$-coefficients, we have

$$
\begin{aligned}
&\sum_j (-\mathsf{v})^j \, \mathrm{H}^j_{c,G}(\bar{\mathcal{Z}}_z)^\vee \\
&= \sum_j (-\mathsf{v})^j \, \mathrm{Hom}^{2N-j}(\bar{K}_{z,G}|_{G^{uni}}, \mathcal{S}_G) && \text{by Lemma 4.1} \\
&= \sum_{i,j} (-\mathsf{v})^j \, \mathrm{Hom}^{2N-j+i}({}^p\mathcal{H}^i(\bar{K}_{z,G})|_{G^{uni}}, \mathcal{S}_G) && \text{by the decomp. thm.} \\
&= \sum_{k,i} (-\mathsf{v})^{k+i} \, \mathrm{Hom}^{2N-k}({}^p\mathcal{H}^i(\bar{K}_{z,G})|_{G^{uni}}, \mathcal{S}_G) \\
&= \sum_{k,i,\chi} (-\mathsf{v})^{k+i} (\mathcal{A}_\chi : {}^p\mathcal{H}^i(\bar{K}_z)) \, \mathrm{Hom}^{4N-\dim G-k}(\mathcal{S}_{\chi,G}, \mathcal{S}_G) && \text{by Lemma 4.3} \\
&= \mathsf{v}^{\ell(z)} \sum_{k,\chi,\psi} (-\mathsf{v})^k \{\chi, \psi\} \psi_{\mathsf{v}}(c_z) \, \mathrm{Hom}^{4N-k}(\mathcal{S}_{\chi,G}, \mathcal{S}_G) && \text{by Theorem 2.1} \\
&= \mathsf{v}^{4N+\ell(z)} \sum_{k,\chi,\psi} (-\mathsf{v})^{-j} \{\chi, \psi\} \psi_{\mathsf{v}}(c_z) \, \mathrm{Hom}^j(\mathcal{S}_{\chi,G}, \mathcal{S}_G) \\
&= \mathsf{v}^{4N+\ell(z)} \sum_i \mathsf{v}^{-2i} \, \mathrm{Sym}^i(\bar{\mathbf{Q}}_\ell \otimes \mathbb{X}) \otimes \sum_{\chi,\psi} \{\chi, \psi\} \psi_{\mathsf{v}}(c_z) V_\chi && \text{by Lemma 4.2.}
\end{aligned}
$$

The trace of w on the last formal series is

$$\mathsf{v}^{4N+\ell(z)} \mathbb{S}_{\mathsf{v}^{-2}}(w) \tau_{z,\mathsf{v}}(w).$$

Let r be the rank of G. To finish the proof, we use the following trick that also appeared in [Tri21]:

$$\mathbb{S}_{\mathsf{q}^{-1}}(w) = \frac{1}{\det(1 - \mathsf{q}^{-1} w \mid \mathbb{X})} = \frac{(-1)^{\ell(w)} \mathsf{q}^r}{\det(1 - \mathsf{q} w \mid \mathbb{X})} = (-1)^{\ell(w)} \mathsf{q}^r \mathbb{S}_{\mathsf{q}}(w).$$

Since $2N + r = \dim G$, we arrive at the desired identity. $\square$

4.2. **Virtual Weight Polynomials.** The results of [Tri21] imply a generalization of Theorem 1.3 at the cost of greater technicalities. We sketch the proof below.

For any scheme X of finite type over $\mathbf{C}$, equipped with a G-action, we define the *G-equivariant virtual weight polynomial* of X as the formal alternating sum

$$\mathrm{E}_G(\mathsf{v}, X) = \sum_i (-1)^i \mathsf{v}^j \, \mathrm{gr}^{\mathsf{W}}_j \, \mathrm{H}^i_{c,G}(X),$$

where $\mathsf{W}_{\leq *}$ denotes the *weight filtration* on $\mathrm{H}^i_{c,G}(X)$ arising from Hodge theory. If W acts on $\mathrm{H}^i_{c,G}(X)$ for all i, then $\mathrm{E}_G(X)$ can be upgraded to a polynomial over the representation ring of W with $\bar{\mathbf{Q}}_\ell$-coefficients. In particular, for $X = \bar{\mathcal{Z}}_z$, the weight filtration coincides with the cohomological filtration, giving

$$\mathrm{E}_G(\bar{\mathcal{Z}}_z) = \sum_i (-\mathsf{v})^i \mathrm{H}^i_{c,G}(\bar{\mathcal{Z}}_z).$$

This follows from the chain of equalities in the proof of Theorem 1.3, computing $\mathrm{H}^i_{c,G}(\bar{\mathcal{Z}}_z)^\vee$ in terms of the spaces $\mathrm{Hom}^*(\mathcal{S}_{\chi,G}, \mathcal{S}_G)$. With respect to the natural

Weil structures on the $\bar{\mathbf{Q}}_\ell$-complexes, these equalities arise from weight-preserving isomorphisms, and as in [RR16], the weight filtration on $\mathrm{Hom}^*(\mathcal{S}_{\chi,G}, \mathcal{S}_G)$ coincides with the filtration by Hom-degree.

The function $X \mapsto \mathrm{E}_G(\mathsf{v}, X)$ factors through the Grothendieck ring of schemes of finite type with G-actions. In particular, if $U \subseteq X$ is an open G-stable subscheme, then $\mathrm{E}_G(\mathsf{v}, X) = \mathrm{E}_G(\mathsf{v}, U) + \mathrm{E}_G(\mathsf{v}, X \setminus U)$. This leads us to consider the open, G-stable subscheme $\mathcal{Z}_z \subseteq \bar{\mathcal{Z}}_z$ defined by

$$\begin{aligned}\mathcal{Z}_z &= Y_e^{uni} \times_G Y_z \\ &\simeq Y_e^{uni} \times_{G^{uni}} Y_z^{uni}.\end{aligned}$$

Observe that $\mathrm{H}_c^*(\mathcal{Z}_z)$ receives a graded $\mathrm{Hom}^*(\mathcal{S}_G, \mathcal{S}_G)$-action, hence a W-action, from an analogue of Lemma 4.1.

In [Tri21], we prove an analogue of Theorem 1.3 for arbitrary z, with $\mathcal{Z}_z$ in place of $\bar{\mathcal{Z}}_z$. Namely, writing $(\delta_w)_w$ for the standard Hecke basis as in Section 1, we have

$$\mathrm{tr}(w \mid \mathrm{E}_G(\mathcal{Z}_z)) = (-1)^{\ell(w)} \mathsf{v}^{2\dim G} \mathbb{S}_{\mathsf{v}^2}(w) \cdot \sum_{\chi,\psi} \{\chi, \psi\} \psi_{\mathsf{v}}(\delta_z) \chi(w).$$

For all $y \leq z$ in the Bruhat order on W, let $p_{y,z}(\mathsf{q}) \in \mathbf{Z}[\mathsf{q}]$ denote the associated Kazhdan–Lusztig polynomial, so that

$$\mathsf{v}^{\ell(z)} c_z = \sum_{y \leq z} p_{y,z}(\mathsf{v}^2) \delta_y.$$

We conclude:

Theorem 4.4. *For any $z \in W$ and $w \in W$, we have*

$$\sum_{y \leq z} p_{y,z}(\mathsf{v}^2)\, \mathrm{tr}(w \mid \mathrm{E}_G(\mathcal{Z}_y)) = (-1)^{\ell(w)} \mathsf{v}^{2\dim G + \ell(z)} \mathbb{S}_{\mathsf{v}^2}(w) \tau_{z,\mathsf{v}}(w),$$

where the virtual W-action on $\mathrm{E}_G(\mathcal{Z}_y)$ arises from the W-action on $\mathrm{H}_{c,G}^*(\mathcal{Z}_y)$: i.e., *from convolution.*

When z is rationally smooth, $p_{y,z}(\mathsf{q}) = 1$ for all $y \leq z$. So in this case, Theorem 4.4 specializes to Theorem 1.3 by the additivity of E_G.

5. Induction Theorems

The goal of this section is to prove Corollary 5.6, a refinement of Proposition 1.9. We keep the hypotheses and notation of Section 2.

We identify W with $N_G(T)/T$ for a fixed maximal torus $T \subseteq G$. Recall that the system of simple reflections $S \subseteq W$ determines and is determined by a choice of Borel $B \subseteq G$ containing T.

Fix a subset $S' \subseteq S$. Let W' be the parabolic subgroup of W generated by S'. Let $H_{W'}$ be the Hecke algebra of W'. The inclusion $S' \subseteq S$ induces an inclusion $H_{W'} \subseteq H_W$.

In what follows, we write $\tau^W_{z,\vee}$, $\tau^{W'}_{z,\vee}$ in place of $\tau_{z,\vee}$, to avoid ambiguity. Let $\mathrm{Ind}^W_{W'}$ denote induction of class functions from W' to W. The following result is proved in [Tri21], as part of a more general compatibility for the linear functions on $H_{W'}$ and H_W defined by $c_z \mapsto \tau^{W'}_{z,\vee}$ and $c_z \mapsto \tau^W_{z,\vee}$.

Theorem 5.1. *For all $z \in W'$, we have*

$$\tau^W_{z,\vee} = \mathrm{Ind}^W_{W'} \tau^{W'}_{z,\vee}$$

as class functions on W.

Let $P = BW'B$, a parabolic subgroup of G. Every parabolic of G arises in this way from some choice of Borel pair $T \subseteq B \subseteq G$ and subset $S' \subseteq S$, since all Borel pairs are conjugate.

Let G' be the Levi factor of P, viewed as a connected algebraic subgroup of G. Let $\mathcal{B}'$ and $(G')^{uni}$ be the flag variety and unipotent locus of G', respectively. For any $u \in (G')^{uni}(\mathbf{k})$, let $spr_{u,G} \colon W \to \mathbf{Z}$ and $spr_{u,G'} \colon W' \to \mathbf{Z}$ be defined by

$$\begin{aligned} spr_{u,G}(w) &= \mathrm{tr}(w \mid \mathrm{H}^*(\mathcal{B}_u)) \\ spr_{u,G'}(w) &= \mathrm{tr}(w \mid \mathrm{H}^*(\mathcal{B}'_u)). \end{aligned}$$

For $\mathbf{k}$ of characteristic zero, Alvis–Lusztig observed that

$$\mathrm{Ind}^W_{W'} spr_{u,G'} = spr_{u,G} \tag{5.1}$$

as class functions on W. Lusztig later gave a proof valid for arbitrary $\mathbf{k}$; for $\mathbf{k} = \mathbf{C}$, Treumann gave a proof via equivariant localization and Rossman's refinement of the Springer action to an action on homotopy types.

Below, we need a refinement of (5.1) to the level of the characters $spr_{\psi,G}$. As it is Treumann's proof that we will refine, we assume throughout the rest of the section that $\mathbf{k} = \mathbf{C}$. I am grateful to Roman Bezrukavnikov for suggesting that the proof of (5.1) via equivariant localization admits such a refinement.

Parallel to our notation in (1.3), let $A'_u = \pi_0(G'_u)$, and for any $\kappa' \in \mathrm{Irr}(A'_u)$, let $spr_{u,\kappa'} \colon W' \to \mathbf{Z}$ be defined by

$$spr_{u,\kappa'}(w) = \sum_i \mathrm{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}'_u)_{\kappa'}).$$

The inclusion $G'_u \subseteq G_u$ descends to a homomorphism $\iota_u \colon A'_u \to A_u$. We write ι_u^* and $\iota_{u,*}$ for the pullback and pushforward of class functions along ι_u, respectively. Explicitly, for any class function θ on A'_u, we have

$$\iota_{u,*}\theta(a) = \mathrm{Ind}^{A_u}_{\iota_u(A'_u)} \mathrm{Avg}_{\iota_u} \theta,$$

where Avg_{ι_u} is in turn defined by

$$(\mathrm{Avg}_{\iota_u} \theta)(a) = \frac{1}{|\ker(\iota_u)|} \sum_{a' \in \iota_u^{-1}(a)} \theta(a'),$$

and decategorifies the operation on finitely-generated A'_u-modules of taking $\ker(\iota_u)$-invariants.

Remark 5.2. We believe that ι_u is always injective. However, we do not know a reference for this statement in the literature, nor a uniform proof that avoids reduction to the case where G is almost-simple. When ι_u is injective, ι_u^* and $\iota_{u,*}$ simplify to the usual restriction and induction of class functions.

For any class function η on A_u, we have the adjunction

$$(\iota_u^*\eta, \theta)_{A'_u} = (\eta, \iota_{u,*}\theta)_{A_u}, \tag{5.2}$$

where $(-,-)_{A_u}$, $(-,-)_{A'_u}$ denote the appropriate scalar products.

Theorem 5.3. *For any $u \in (G')^{uni}(\mathbf{k})$ and $\kappa' \in \mathrm{Irr}(A'_u)$, we have*

$$\mathrm{Ind}_{W'}^{W} spr_{u,\kappa'} = \sum_{\kappa \in \mathrm{Irr}(A_u)} (\kappa, \iota_{u,*}\kappa')_{A_u}\, spr_{u,\kappa}$$

as class functions on W.

Example 5.4. Applying $\iota_{u,*}$ to the character of the regular representation of A'_u produces the character of the regular representation of A_u. From this fact, and the linearity of $\mathrm{Ind}_{W'}^{W}$ and $\iota_{u,*}$, we see that Theorem 5.3 implies (5.1).

Example 5.5. Take G and G' of types B_3 and B_2, respectively. The embedding of G' into G is unique up to conjugacy. The unipotent classes of G, *resp.* G', are labeled by certain integer partitions of 7, *resp.* 5 in [Car93] or CHEVIE. The irreducible characters of W, *resp.* W', are labeled by the integer bipartitions of 3, *resp.* 2.

Let u belong to the unipotent conjugacy class in G' labeled 311. Its conjugacy class in G is labeled 31111. The groups A_u, A'_u are both cyclic of order 2. Writing ε, ε' for their respective nontrivial characters, we have the following decompositions of the characters $spr_{u,\kappa}$, $spr_{u,\kappa'}$:

	$spr_{u,1_{A_u}}$	$spr_{u,\varepsilon}$
3.	1	
21.	1	1
2.1	2	
.3		
1.2	1	
11.1	1	1
111.		1
1.11	1	
.21		
.111		

	$spr_{u,1_{A'_u}}$	$spr_{u,\varepsilon'}$
2.	1	
11.		1
1.1	1	
.2		
.11		

The map $\iota_u \colon A'_u \to A_u$ is an isomorphism. So here, Theorem 5.3 predicts that

$$\mathrm{Ind}_{W'}^{W} spr_{u,1_{A'_u}} = spr_{u,1_{A_u}} \quad \text{and} \quad \mathrm{Ind}_{W'}^{W} spr_{u,\varepsilon'} = spr_{u,\varepsilon}.$$

Indeed, the above identities are in agreement with the following table of the scalar products $(\chi, \mathrm{Ind}_{W'}^{W} \chi')_W$ for $\chi \in \mathrm{Irr}(W)$ and $\chi' \in \mathrm{Irr}(W')$:

	2.	11.	1.1
3.	1		
21.	1	1	
2.1	1		1
.3			
1.2			1
11.1		1	1
111.		1	
1.11			1
.21			
.111			

Proof of Theorem 5.3. We will merely sketch what adjustments are needed to the proof of Theorem 1.1 in [Tre09, 15–16].

Note that Treumann's L, ν, $(-)^{\mathrm{rss}}$ are our G', u, $(-)^{rs}$. In particular, his W_G, W_L are our W, W'. Also, he works with the topological spaces formed by the $\mathbf{C}$-points of his algebraic varieties, in the analytic topology, rather than the varieties themselves in the étale topology. Via the comparison theorem of [AGV+73, Exposé XVI], his conclusions for singular cohomology can be transported to étale cohomology, once we extend scalars from $\mathbf{Q}$ to $\bar{\mathbf{Q}}_\ell$.

As in [Tre09], let U(1) be the compact real form of GL_1, and fix an embedding of U(1) into G such that the neutral component of its centralizer is precisely G'. The main structure in the proof is a commutative square of topological spaces, which in our notation becomes:

$$\begin{tikzcd} E^{rs}_{u,G'} \arrow[r] \arrow[d, "\chi'"'] & E^{rs}_{u,G} \arrow[d, "\chi"] \\ B(\delta, T /\!\!/ W')^{rs} \arrow[r, "\iota"] & B(\delta, T /\!\!/ W)^{rs} \end{tikzcd}$$

Above: The notation $B(\delta, -)$ denotes an open ball of radius $\delta > 0$ around a chosen point. The bottom horizontal arrow is a fiber bundle with fiber W/W'. The vertical arrows are Ehresmann fibrations: that is, smooth, proper, surjective maps. The fibers of χ', *resp.* χ, are homotopy equivalent to $\mathcal{B}_u$, *resp.* $\mathcal{B}'_u$. There is a U(1)-action on $E^{rs}_{u,G}$, along which χ is invariant. The top horizontal arrow is an embedding that identifies $E^{rs}_{u,G'}$ with the set of U(1)-fixed points of $E_{u,G}$.

Let $\underline{\mathbf{Q}}_{(-),\mathrm{U}(1)}$ denote the U(1)-equivariant constant sheaf with $\mathbf{Q}$-coefficients over a U(1)-space. The monodromy action of the fundamental group of $B(\delta, T /\!\!/ W)^{rs}$, *resp.* $B(\delta, T /\!\!/ W')^{rs}$, on the stalks of $\underline{\mathbf{Q}}_{E^{rs}_{u,G},\mathrm{U}(1)}$, *resp.* $\underline{\mathbf{Q}}_{E^{rs}_{u,G'},\mathrm{U}(1)}$, factors through W, *resp.* W'. This induces the Springer action on $\mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}_u, \mathbf{Q})$, *resp.* $\mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}'_u, \mathbf{Q})$, with the same sign convention as in Section 4.

As in [Tre09], we identify $\mathrm{H}^*_{\mathrm{U}(1)}(pt, \mathbf{Q})$ with $\mathbf{Q}[t]$, where $\deg t = 2$. The main input in Treumann's proof is the localization theorem for U(1)-equivariant cohomology, showing that in the U(1)-equivariant derived category of sheaves in the analytic topology on $B(\delta, T /\!\!/ W)^{rs}$, we have a W-equivariant isomorphism:

$$\iota_* \chi'_* \underline{\mathbf{Q}}_{E^{rs}_{u,G'}, \mathrm{U}(1)} \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}] \simeq \chi_* \underline{\mathbf{Q}}_{E^{rs}_{u,G}, \mathrm{U}(1)} \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}]. \tag{5.3}$$

This induces a W-equivariant isomorphism:

$$\mathrm{Ind}^W_{W'} \mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}'_u, \mathbf{Q}) \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}] \simeq \mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}_u, \mathbf{Q}) \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}]. \tag{5.4}$$

The remainder of Treumann's proof amounts to showing the degeneration of a spectral sequence, which in turn shows how ordinary cohomology can be recovered from equivariant cohomology after collapsing cohomological degree.

Since the copy of $\mathrm{U}(1)$ in G commutes with G' by construction, we see that the G'_u-actions on $\mathcal{B}_u$ and $\mathcal{B}'_u$ induce A'_u-actions on $\mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}_u, \mathbf{Q})$ and $\mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}'_u, \mathbf{Q})$. These in turn induce A'_u-actions on $\underline{\mathbf{Q}}_{E^{rs}_{u,G}, \mathrm{U}(1)}$ and $\underline{\mathbf{Q}}_{E^{rs}_{u,G'}, \mathrm{U}(1)}$. Since $E^{rs}_{u,G'}$ is just the $\mathrm{U}(1)$-fixed locus of $E_{u,G}$, the isomorphism (5.3) is equivariant with respect to these actions. So (5.4) is also A'_u-equivariant. So for any $\kappa' \in \mathrm{Irr}(A'_u)$, we have

$$\mathrm{Ind}^W_{W'} \mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}'_u, \mathbf{Q})_{\kappa'} \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}] \simeq \mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}_u, \mathbf{Q})_{\kappa'} \otimes_{\mathbf{Q}[t]} \mathbf{Q}[t^{\pm 1}],$$

where, on the right-hand side, we implicitly pull back $\mathrm{H}^*_{\mathrm{U}(1)}(\mathcal{B}_u, \mathbf{Q})$ along ι_u to a representation of A'_u. The spectral-sequence degeneration is compatible with this refinement to κ'-multiplicity spaces. Altogether,

$$\mathrm{Ind}^W_{W'} \mathit{spr}_{u,\kappa'} = \sum_{\kappa \in \mathrm{Irr}(A_u)} (\iota_u^* \kappa, \kappa')_{A'_u} \mathit{spr}_{u,\kappa},$$

which is equivalent to the theorem statement via (5.2). □

In what follows, let Υ_G be the set of G-conjugacy classes of pairs (u, κ) such that $u \in G^{uni}(\mathbf{k})$ and $\kappa \in \mathrm{Irr}(A_u)$. Let $\Upsilon_{G'}$ be defined similarly, with G replaced by G'. For fixed $u \in G^{uni}(\mathbf{k})$, let $\Upsilon_{G',u} \subseteq \Upsilon_{G'}$ be the subset of classes $[u', \kappa']$ such that u' becomes conjugate to u in G.

When $[u, \kappa] \in \Upsilon_G$ is the image of $\psi \in \mathrm{Irr}(W)$ under the Springer correspondence, we set $\alpha_{u,\kappa} = \alpha_{\psi,G}$. For $[u', \kappa'] \in \Upsilon_{G'}$, we define $\alpha_{u',\kappa'}$ similarly. With this notation, we can state a refinement of Proposition 1.9.

Corollary 5.6. *As above, let $W' \subseteq W$ be a parabolic subgroup and $G' \subseteq G$ the corresponding Levi relative to a fixed Borel pair of G. If $z \in W'$, then*

$$\alpha^z_{u,\kappa} = \sum_{[u',\kappa'] \in \Upsilon_{G',u}} (\kappa, \iota_{u',*}\kappa')_{A_u} \alpha^z_{u',\kappa'} \quad \textit{for all } [u, \kappa] \in \Upsilon_G.$$

In particular, the $\alpha^z_{\psi,G}$ for $\psi \in \mathrm{Irr}(W)$ are nonnegative integer sums of the $\alpha^z_{\psi',G'}$ for $\psi' \in \mathrm{Irr}(W')$.

Proof. We have

$$\begin{aligned}
\tau^W_{z,\mathsf{v}} &= \mathrm{Ind}^W_{W'} \tau^{W'}_{z,\mathsf{v}} && \text{by Theorem 5.1}\\
&= \sum_{[u',\kappa'] \in \Upsilon_{G'}} \alpha^z_{u',\kappa'} \, \mathrm{Ind}^W_{W'} \, spr_{u',\kappa'}\\
&= \sum_{[u',\kappa'] \in \Upsilon_{G'}} \alpha^z_{u',\kappa'} \sum_{\kappa \in \mathrm{Irr}(A_{u'})} (\kappa, \iota_{u',*}\kappa')_{A_u} \, spr_{u,\kappa} && \text{by Theorem 5.3}\\
&= \sum_{[u] \in G^{uni}/G} \sum_{[u',\kappa'] \in \Upsilon_{G',u}} \alpha^z_{u',\kappa'} \sum_{\kappa \in \mathrm{Irr}(A_u)} (\kappa, \iota_{u',*}\kappa')_{A_u} \, spr_{u,\kappa}\\
&= \sum_{[u,\kappa] \in \Upsilon_G} \left(\sum_{[u',\kappa'] \in \Upsilon_{G',u}} (\kappa, \iota_{u',*}\kappa')_{A_u} \alpha^z_{u',\kappa'} \right) spr_{u,\kappa}.
\end{aligned}$$

Since the decomposition of $\tau^W_{z,\mathsf{v}}$ into the characters $spr_{u,\kappa}$ is unique, we deduce the desired claim. □

6. The Cell Order

In this section, we first review two-sided cells following [KL79] and [Lus84, Chapter 5], then prove Proposition 1.10, Corollary 6.4, and some other facts.

Recall that H_W denotes the Hecke algebra of W. For all $x, y \in W$, write $y \xrightarrow{L} x$, *resp.* $y \xrightarrow{R} x$, to mean that there exists some $\alpha \in H_W$ such that c_y has nonzero coefficient when αc_x, *resp.* $c_x \alpha$, is expanded in the Kazhdan–Lusztig basis. Write $y \xrightarrow{LR} x$ to mean that either $y \xrightarrow{L} x$ or $y \xrightarrow{R} x$. The relation $\xrightarrow{LR}$ is reflexive, so its transitive closure $\leq_{LR}$ is a partial order on W. The identity e is the unique topmost element in $\leq_{LR}$, while the longest element $w_\circ$ is the bottommost.

Write $x \sim_{LR} y$ to mean that both $y \leq_{LR} x$ and $x \leq_{LR} y$. Then $\sim_{LR}$ is an equivalence relation on W. The *two-sided cells* of W are the equivalence classes under $\sim_{LR}$. The partial order $\leq_{LR}$ descends to a partial order $\leq$ on the set of two-sided cells C_W. The singleton $\{e\}$ is the unique topmost two-sided cell, while the singleton $\{w_\circ\}$ is the bottommost.

For any two-sided cell $\mathcal{F} \in \mathsf{C}_W$, let $M_{\leq \mathcal{F}}$, *resp.* $M_{< \mathcal{F}}$, be the $\mathbf{Z}[\mathsf{v}^{\pm 1}]$-submodule of H_W spanned by the elements c_w with $w \in \mathcal{F}'$ for some $\mathcal{F}' \leq \mathcal{F}$, *resp.* $\mathcal{F}' < \mathcal{F}$. The definition of $\leq_{LR}$ shows that $M_{\leq \mathcal{F}}$ and $M_{< \mathcal{F}}$ are, in fact, two-sided ideals of H_W. We define the *two-sided cell module* associated with $\mathcal{F}$ to be the left H_W-module formed by $M_{\mathcal{F}} := M_{\leq \mathcal{F}} / M_{< \mathcal{F}}$.

Each simple module over $\mathbf{Q}(\mathsf{v}) \otimes H_W$ occurs in the two-sided cell module $M_{\mathcal{F}}$ for a unique $\mathcal{F} \in \mathsf{C}_W$. As any such module has a character of the form χ_{v} for some $\chi \in \mathrm{Irr}(W)$, we obtain a map $\mathrm{Irr}(W) \to \mathsf{C}_W$, which we will denote by $\chi \mapsto \mathcal{F}(\chi)$. By work of Barbasch–Vogan [Lus84, §5.15], the pairing $\{-,-\}$ reviewed in Appendix A is block-diagonal with respect to the fibers of this map. That is:

Lemma 6.1 (Barbasch–Vogan)**.** *If $\mathcal{F}(\chi) \neq \mathcal{F}(\psi)$, then $\{\chi, \psi\} = 0$.*

The sets $\mathrm{Irr}(W)_{\mathcal{F}} = \{\chi \in \mathrm{Irr}(W) \mid \mathcal{F}(\chi) = \mathcal{F}\}$ may be called the *families* of the irreducible characters of W. Lusztig observed in [Lus97] that under the Springer correspondence, the partition of $\mathrm{Irr}(W)$ into families $\mathrm{Irr}(W)_{\mathcal{F}}$ corresponds to a partition of G^{uni} into subvarieties $G^{uni}_{\mathcal{F}}$, now called the *pieces* of G^{uni}. In particular, if $\prec_{uni}$, *resp.* $\prec_{LR}$, denotes the partial order on $\mathrm{Irr}(W)$ obtained by pulling back the reverse of the closure order on the unipotent conjugacy classes of G, *resp.* pulling back the partial order on two-sided cells of W, then:

Lemma 6.2 (Lusztig)**.** *$\prec_{uni}$ is a refinement of $\prec_{LR}$.*

Proposition 1.10 is part (2) below.

Proposition 6.3. *Let $\mathcal{F} \in \mathsf{C}_W$ and $z \in \mathcal{F}$.*

(1) *If $\chi \in \mathrm{Irr}(W)$ and $\mathcal{F} < \mathcal{F}(\chi)$, then*

$$\sum_{\chi,\psi} \{\chi, \psi\}\psi_{\mathsf{v}}(c_z) = 0.$$

(2) *If $\psi \in \mathrm{Irr}(W)$ and $\mathcal{F} < \mathcal{F}(\psi)$, then $\alpha^z_{\psi,G} = 0$.*

Proof. (1) Fix $\psi \in \mathrm{Irr}(W)$. If $\{\chi, \psi\} \neq 0$, then $\mathcal{F}(\psi) = \mathcal{F}(\chi)$ by Lemma 6.1, from which $\mathcal{F} < \mathcal{F}(\psi)$, which in turn forces $\psi_{\mathsf{v}}(c_z) = 0$ by the definition of $\leq_{LR}$. So every term vanishes in the double sum above.

(2) Observe that the matrix of scalar products $((\chi, spr_\psi)_W)_{\chi,\psi}$ is unipotent upper-triangular in a total order on $\mathrm{Irr}(W)$ refining $\prec_{uni}$. Hence its inverse is also upper-triangular in the same total order. Now, the claim follows from combining (1) and Lemma 6.2. ☐

Let 1_W be the trivial character of W; let ε be the sign character defined by $\varepsilon(w) = (-1)^{\ell(w)}$. Corollary 1.11 is parts (2)–(3) below.

Corollary 6.4. (1) *If $z \neq e$, then $\alpha^z_{\varepsilon,G} = 0$.*

(2) *If $\psi \neq 1_W$, then $\alpha^{w_\circ}_{\psi,G} = 0$.*

(3) *We have*

$$\alpha^{w_\circ}_{1_W,G}(\mathsf{v}) = \sum_{w \in W} \mathsf{v}^{2\ell(w)-\ell(w_\circ)}.$$

Proof. (1) This follows from observing that the module for the topmost two-sided cell $\{e\}$ is simple with character ε_{v}: in other words, that $\mathrm{Irr}(W)_{\{e\}} = \{\varepsilon\}$. Indeed, ε_{v} is determined by the property that $\varepsilon_{\mathsf{v}}(\delta_s) = -1$ for all simple reflections s. This means $\varepsilon_{\mathsf{v}}(c_s) = 0$ for all s, which in turn describes the character of the cell module.

(2) This follows from observing that the module for the bottommost two-sided cell $\{w_\circ\}$ is simple with character $1_{W,\mathsf{v}}$: in other words, that $\mathrm{Irr}(W)_{\{w_\circ\}} = \{1_W\}$. This in turn follows from (1) and [Lus84, Lemma 5.14(iii)].

(3) By (2), we must have $\tau_{w_\circ,\mathsf{v}} = \alpha^{w_\circ}_{1_W,G} spr_{1_W}$. But $spr_{1_W} = 1_W$, so we deduce that $\alpha^{w_\circ}_{1_W,G} = (1_W, \tau_{w_\circ,\mathsf{v}})_W = 1_{W,\mathsf{v}}(c_{w_\circ})$. Since $\mathsf{v}^{\ell(w_\circ)} c_{w_\circ} = \sum_{w\in W} \delta_w$ and $1_{W,\mathsf{v}}(\delta_w) = \mathsf{v}^{2\ell(w)}$, we win. ☐

7. Examples

Recall $(\triangle)^z_{\psi,G}, (\pm)^z_{\psi,G}, (+)^z_{\psi,G}$ from §1.3. In this section, we summarize their behavior in each irreducible root system outside type A of rank ≤ 5, as well as in those of rank 6 when z is rationally smooth.

Recall that in [Hai93], Haiman verified Conjecture A—stating that $(\triangle)^z_{\psi,G}$ and $(+)^z_{\psi,G}$ hold in type A—up through rank 6.

7.1. **Methodology.** We can rewrite identity (1.4) as the collection of identities

$$(\chi, \tau_{z,\mathsf{v}})_W = \sum_{\psi \in \mathrm{Irr}(W)} \alpha^z_{\psi,G}(\mathsf{v})(\chi, spr_{\psi,G})_W \quad \text{for } \chi \in \mathrm{Irr}(W).$$

Explicitly, the left-hand side is $\sum_\psi \{\chi, \psi\}\psi_{\mathsf{v}}(c_z)$. In the CHEVIE package for GAP3:

- The pairings $\{\chi, \psi\}$ can be computed from the functions `UnipotentCharacter` and `AlmostCharacter`.
- The values $\psi_{\mathsf{v}}(c_z)$ can be computed from `HeckeCharValues`.
- The pairings $(\chi, spr_{\psi,G})_W$ can be computed from `ICCTable`. In the GAP3 manual and other literature on the Lusztig–Shoji algorithm, $(\chi, spr_{\psi,G})_W$ is denoted $P_{\psi,\chi}$: See [GM99, Ach21, Ach09].

Note that the ordering of $\mathrm{Irr}(W)$ used in `ICCTable` differs from the ordering used in the other functions above. After matching the orderings, we obtain the $\alpha^z_{\psi,G}$ by multiplying the matrix $((\chi, \tau_{z,\mathsf{v}})_W)_{\chi,z}$ on the left by the inverse of the matrix $((\chi, spr_{\psi,G})_W)_{\chi,\psi}$. To process the CHEVIE data, we used Python scripts written with help from Claude Sonnet 4.6. The data and scripts are available at:

https://mqtrinh.github.io/math/research/code/chevie/

The webpage also includes data for the root systems of types B_7, C_7, D_7 when z is rationally smooth of length ≤ 24.

7.2. **Notation.** We use the labels in CHEVIE, in the ordering of `ICCTable`, for irreducible characters of W. We use reduced words in the simple reflections $s_i \in S$ to label elements of W, and use the numbering in CHEVIE to label the s_i.[6] We write $w_\circ$ for the longest element.

In rank ≤ 3, we provide tables of the $\alpha^z_{\psi,G}$, listing $\psi \in \mathrm{Irr}(W)$ along rows and $z \in W$ along columns. To compactify the entries, we write

$$(\mathsf{a}_m \cdots \mathsf{a}_1\mathsf{a}_0) := \mathsf{a}_m\mathsf{v}^m + \cdots + \mathsf{a}_1\mathsf{v} + \mathsf{a}_0 + \mathsf{a}_1\mathsf{v}^{-1} + \cdots + \mathsf{a}_m\mathsf{v}^{-m}.$$

In all ranks, we provide tables stating for each ψ whether $(\triangle)^z_{\psi,G}, (\pm)^z_{\psi,G}, (+)^z_{\psi,G}$ fail for some z and whether ψ is inflated from certain quotients. Here, we also give the label in CHEVIE for the image of ψ under the Springer correspondence.

For any $w = s_{i_1} \cdots s_{i_\ell} \in W$, we write $\varphi_{i_1 \cdots i_\ell}$ to denote the quotient map out of W that is reduction modulo w. We set $\varphi_\circ = \varphi_{w_\circ}$.

[6] See https://webusers.imj-prg.fr/~jean.michel/gap3/htm/chap085.htm.

7.3. Type B_2. The $\alpha^z_{\psi,G}$:

	e	2	1	21, 12	212	121	$w_\circ$
2.				(101)	(1020)	(1020)	(10202)
11.			(10)	1	(10)		
1.1			(10)			$-$(10)	
.2		(10)		1		(10)	
.11	1						

The unique maximal quotient map to a product of symmetric groups is $\varphi_\circ\colon W \to W(A_1 \times A_1)$.

		¬(△)	¬(±)	¬(+)	$\varphi_\circ$-inflation
2.	5				✓
11.	311(11)				✓
1.1	311			✓	
.2	221				✓
.11	11111				✓

7.4. Type G_2. The $\alpha^z_{\psi,G}$:

	e	2	1	21, 12	212	121	2121, 1212	21212	12121	$w_\circ$
$\phi_{1,0}$				(101)	(1020)	(1020)	(10202)	(102020)	(102020)	(1020202)
$\phi'_{1,3}$							1	(10)	(10)	
$\phi_{2,1}$				-1	$-$(10)	$-$(10)	-1			
$\phi_{2,2}$		(10)		1		(10)			$-$(10)	
$\phi''_{1,3}$			(10)	1	(10)		1		(10)	
$\phi_{1,6}$	1									

The maximal quotient maps to products of symmetric groups are $\varphi_{1212}\colon W \to W(A_1 \times A_1)$ and $\varphi_\circ\colon W \to W(A_2)$. The latter is what we called the G_2-to-A_2 quotient in §1.4.

		¬(△)	¬(±)	¬(+)	φ_{1212}-inflation	$\varphi_\circ$-inflation
$\phi_{1,0}$	G_2				✓	✓
$\phi'_{1,3}$	$G_2(a_1)(21)$				✓	
$\phi_{2,1}$	$G_2(a_1)$			✓		
$\phi_{2,2}$	$\tilde{A}_1$			✓		✓
$\phi''_{1,3}$	A_1				✓	
$\phi_{1,6}$	1				✓	✓

The failure of $(+)^z_{\psi,G}$ for $\psi = \phi_{2,2}$ is the reason for the complication involving the G_2-to-A_2 quotient in Conjecture B.

7.5. Type B_3. The $\alpha^z_{\psi,G}$:

	e	3, 2	1	32, 23	21, 12	13	212	321, 213, 132, 123	232	121	3212, 2123
3.								(1010)			(10202)
21.					(101)		(1020)	(10)		(1020)	(203)
2.1					(101)		(1020)			(1020)	(101)
.3				(101)				(10)	(1020)		1
1.2				(101)				(10)	(1020)		1
11.1						(102)		(10)			
111.			(10)				(10)				1
1.11			(10)							$-$(10)	
.21		(10)		1						(10)	
.111	1										

	2321, 1321, 1213, 1232	2132	1212	32123	21232, 23212	21321, 12132	13212, 12123
3.	(10202)	(10202)		(103040)	(103040)	(103030)	(102020)
21.	(102)	(102)	(10202)	(2050)	(1030)	(1030)	(1020)
2.1			(10202)	(1020)		(1020)	(1020)
.3	(102)	(102)		(10)	(10)	(20)	(10)
1.2	(102)	(102)		(10)	(10)	(10)	
11.1		(102)				(10)	
111.				(10)			
1.11							
.21							
.111							

	12321	232123	212321, 123212	213212, 121232	132123	121321	2123212
3.	(103040)	(1030506)	(1040606)	(1030404)	(1030404)	(1030404)	(10407080)
21.	(1030)	(10304)	(204)	(102)	(10304)	(102)	(1030)
2.1			(102)	(102)	(10304)	(102)	(1030)
.3	(1030)		(204)	(102)	(102)	(102)	(2050)
1.2	(1030)		(102)				(1020)
11.1						(102)	
111.							
$\vdots$							

	2132123, 1232123	1212321, 1213212	21232123	12123212	12132123	$w_\circ$
3.	(10407080)	(10305060)	(103050606)	(103050708)	(104080[11]0[12])	(1030507080)
21.	(1030)	(10)			(102)	
2.1	(1030)	(10)	(10304)		(102)	
.3	(1030)	(1020)		(10202)	(10304)	
1.2			(10304)			
11.1						
111.						
$\vdots$						

The unique maximal quotient map to a product of symmetric groups is $\varphi_{1212} \colon W \to W(A_1 \times A_2)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
3.	7				✓
21.	511(11)				✓
2.1	511				
.3	331(11)				✓
1.2	331				
11.1	322				
111.	31111(11)				✓
1.11	31111			✓	
.21	22111				✓
.111	1111111				✓

7.6. **Type C_3.** The $\alpha^z_{\psi,G}$:

	e	3, 2	1	32, 23	21, 12	13	212	321, 213, 132, 123	232	121	3212, 2123
3.								(1010)			(10202)
.3											
2.1								$-(10)$			$-(102)$
1.2				(101)				(10)	(1020)		1
21.					(101)		(1020)	(10)		(1020)	(203)
11.1						(102)		(10)			
.21		(10)		1	1					(10)	
1.11		(10)		1			$-(10)$				-1
111.			(10)		1		(10)				1
.111	1										

	2321, 1321, 1213, 1232	2132	1212	32123	21232, 23212	21321, 12132	13212, 12123
3.	(10202)	(10202)		(103040)	(103040)	(103030)	(102020)
.3						(10)	(10)
2.1	−(102)	−(102)		−(1030)	−(1030)	−(10)	
1.2	(102)	(102)		(10)	(10)	(10)	
21.	(102)	(102)	(10202)	(2050)	(1030)	(1030)	(1020)
11.1		(102)				(10)	
.21							
1.11				−(10)			
111.				(10)			
.111							

	12321	232123	212321, 123212	213212, 121232	132123	121321	2123212
3.	(103040)	(1030506)	(1040606)	(1030404)	(1030404)	(1030404)	(10407080)
.3			(102)	(102)	(102)	(102)	(1030)
2.1	−(1030)	−(10304)	−(102)				
1.2	(1030)		(102)				(1020)
21.	(1030)	(10304)	(204)	(102)	(10304)	(102)	(1030)
11.1						(102)	
.21							
1.11							
111.							
.111							

	2132123, 1232123	1212321, 1213212	21232123	12123212	12132123	$w_\circ$
3.	(10407080)	(10305060)	(103050606)	(103050708)	(104080[11]0[12])	(1030507080)
.3	(1030)	(1020)	(10304)	(10202)	(10304)	
2.1			(10304)			
1.2						
21.	(1030)	(10)			(102)	
11.1						
.21						
⋮						

As in type B_3, it suffices to consider $\varphi_{1212}\colon W \to W(A_1 \times A_2)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
3.	6				✓
.3	42(11)				✓
2.1	42			✓	
1.2	33				
21.	411				✓
11.1	222				
.21	2211(11)				✓
1.11	2211			✓	
111.	21111				✓
.111	111111				✓

7.7. **Type B_4.** The unique maximal quotient map to a product of symmetric groups is $\varphi_{1212}\colon W \to W(A_1 \times A_3)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
4.	9				✓
31.	711(11)				✓
3.1	711				
.4	531(11, 11)				✓
22.	531(2, 11)				✓
2.2	531			✓	
1.3	441				
21.1	522				
211.	51111(11)				✓
2.11	51111				
11.2	333				
.31	33111(11)				✓

1.21	33111				
111.1	32211(11)				
11.11	32211			✓	
.22	22221				✓
1111.	3111111(11)				✓
1.111	3111111			✓	
.211	2211111				✓
.1111	111111111				✓

7.8. **Type C_4.** As in type B_4, it suffices to consider $\varphi_{1212}\colon W \to W(A_1 \times A_3)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
4.	8				✓
.4	62(11)				✓
3.1	62	✓	✓	✓	
1.3	44(11)				
2.2	44				
31.	611				✓
22.	422(11)				✓
21.1	422				
.31	4211(11)				✓
2.11	4211			✓	
11.2	332				
1.21	3311				
.22	2222(11)				✓
11.11	2222			✓	
211.	41111				✓
111.1	22211				
.211	221111(11)				✓
1.111	221111			✓	
1111.	2111111				✓
.1111	11111111				✓

7.9. **Type D_4.** The maximal quotient maps to products of symmetic groups are $\varphi_{12}, \varphi_{14}, \varphi_{24}\colon W \to W(A_3)$. The triality of D_4 permutes these maps transitively, so it is enough to work with φ_{12}.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{12}-inflation
.4	71				✓
1.3	53			✓	
.31	5111				✓
$2 \cdot -$	$44 \cdot +$				
$2 \cdot +$	$44 \cdot -$				
11.2	3311(11)				
1.21	3311			✓	
.22	3221				✓
.211	311111				✓
$11 \cdot -$	$2222 \cdot +$				
$11 \cdot +$	$2222 \cdot -$				
1.111	221111				
.1111	11111111				✓

7.10. **Type F_4.** The unique maximal quotient map to a product of symmetric groups is $\varphi_{2323}\colon W \to W(A_2 \times A_2)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{2323}-inflation
$\phi_{1,0}$	F_4				✓
$\phi'_{2,4}$	$F_4(a_1)(11)$				✓
$\phi_{4,1}$	$F_4(a_1)$	✓	✓	✓	
$\phi''_{2,4}$	$F_4(a_2)(11)$				✓
$\phi_{9,2}$	$F_4(a_2)$	✓	✓	✓	
$\phi'_{8,3}$	C_3				
$\phi''_{8,3}$	B_3				
$\phi'_{1,12}$	$F_4(a_3)(211)$				✓
$\phi''_{6,6}$	$F_4(a_3)(22)$				

$\phi'_{9,6}$	$F_4(a_3)(31)$			✓	
$\phi_{12,4}$	$F_4(a_3)$		✓	✓	
$\phi'_{4,7}$	$C_3(a_1)(11)$			✓	
$\phi_{16,5}$	$C_3(a_1)$			✓	
$\phi'_{6,6}$	$\tilde{A}_2 + A_1$			✓	
$\phi_{4,8}$	$B_2(11)$				✓
$\phi''_{9,6}$	B_2			✓	
$\phi''_{4,7}$	$A_2 + \tilde{A}_1$			✓	
$\phi'_{8,9}$	$\tilde{A}_2$				
$\phi''_{1,12}$	$A_2(11)$				✓
$\phi''_{8,9}$	A_2				
$\phi_{9,10}$	$A_1 + \tilde{A}_1$				
$\phi'_{2,16}$	$\tilde{A}_1(11)$				✓
$\phi_{4,13}$	$\tilde{A}_1$			✓	
$\phi''_{2,16}$	A_1				✓
$\phi_{1,24}$	1				✓

7.11. **Type B_5.** The unique maximal quotient map to a product of symmetric groups is $\varphi_{1212}\colon W \to W(A_1 \times A_4)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
5.	[11]				✓
41.	911(11)				✓
4.1	911				
.5	731(11, 11)				✓
32.	731(2, 11)				✓
3.2	731	✓	✓	✓	
1.4	551(11)				
2.3	551				
31.1	722				
311.	71111(11)				✓
3.11	71111				
22.1	533(11)				
21.2	533			✓	
221.	53111(11, 11)				✓
.41	53111(11, 2)				✓
2.21	53111			✓	
11.3	443				
1.31	44111				
211.1	52211(11)				
21.11	52211				
111.2	33311(11)				
11.21	33311			✓	
.32	33221(11)				✓
1.22	33221				
2111.	5111111(11)				✓
2.111	5111111				
.311	3311111(11)				✓
1.211	3311111				
111.11	3222				
1111.1	3221111(11)				
11.111	3221111			✓	
.221	2222111				✓
11111.	311111111(11)				✓
1.1111	311111111			✓	
.2111	221111111				✓
.11111	11111111111				✓

7.12. **Type C_5.** As in type B_5, it suffices to consider $\varphi_{1212}\colon W \to W(A_1 \times A_4)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	φ_{1212}-inflation
5.	[10]				✓
.5	82(11)				✓
4.1	82	✓	✓	✓	
1.4	64(11)				
3.2	64	✓	✓	✓	

41.	811				$\checkmark$
2.3	55				
32.	622(11)				$\checkmark$
31.1	622				
.41	6211(11)				$\checkmark$
3.11	6211	$\checkmark$	$\checkmark$	$\checkmark$	
11.3	442(11)				
21.2	442			$\checkmark$	
1.31	4411(11)				
2.21	4411				
22.1	433				
.32	4222(11)				$\checkmark$
21.11	4222			$\checkmark$	
311.	61111				$\checkmark$
1.22	3322(11)				
11.21	3322			$\checkmark$	
221.	42211(11)				$\checkmark$
211.1	42211				
111.2	33211				
.311	421111(11)				$\checkmark$
2.111	421111			$\checkmark$	
1.211	331111				
111.11	22222				
.221	222211(11)				$\checkmark$
11.111	222211			$\checkmark$	
2111.	4111111				$\checkmark$
1111.1	2221111				
.2111	22111111(11)				$\checkmark$
1.1111	22111111			$\checkmark$	
11111.	211111111				$\checkmark$
.11111	1111111111				$\checkmark$

7.13. **Type D_5.** The unique maximal quotient map to a product of symmetric groups is $\varphi_{12} \colon W \to W(A_4)$.

		$\neg(\triangle)$	$\neg(\pm)$	$\neg(+)$	φ_{12}-inflation
.5	91				$\checkmark$
1.4	73	$\checkmark$	$\checkmark$	$\checkmark$	
.41	7111				$\checkmark$
2.3	55				
11.3	5311(11)			$\checkmark$	
1.31	5311	$\checkmark$	$\checkmark$	$\checkmark$	
2.21	4411				
.32	5221				$\checkmark$
1.22	3331				
.311	511111				$\checkmark$
11.21	3322				
111.2	331111(11)				
1.211	331111			$\checkmark$	
.221	322111				$\checkmark$
11.111	222211				
.2111	31111111				$\checkmark$
1.1111	22111111				
.11111	1111111111				$\checkmark$

7.14. **Type B_6, Rationally Smooth z.** The unique maximal quotient map to a product of symmetic groups is $\varphi_{1212} \colon W \to W(A_1 \times A_5)$.

There are 65 irreducible characters of W, indexed by the bipartitions of 6. Of these, 22 are inflated along φ_{1212}, being those indexed by the bipartitions (ξ, η) in which one of ξ or η is empty, *cf.* Example 1.13.

Properties $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ always hold when z is rationally smooth. The ψ where $(+)^z_{\psi,G}$ fails for some rationally smooth z are listed below.

4.2	931
1.5	751(11, 11)
3.3	751
3.21	73111
221.1	53311(11, 11)
211.2	53311(11, 2)
21.21	53311
11.31	44311
2.22	53221
2.211	5311111
11.211	3331111
111.111	3222211
11.1111	322111111
1.11111	31111111111

7.15. **Type C_6, Rationally Smooth z.** As in type B_6, it suffices to consider $\varphi_{1212}\colon W \to W(A_1 \times A_5)$, and the 22 irreducible characters inflated along φ_{1212} are those indexed by the bipartitions (ξ, η) in which one of ξ or η is empty.

Properties $(\triangle)^z_{\psi,G}$ and $(\pm)^z_{\psi,G}$ always hold when z is rationally smooth. The ψ where $(+)^z_{\psi,G}$ fails for some rationally smooth z are listed below.

5.1	[10]2
1.5	84(11)
4.2	84
4.11	8211
31.2	642
1.41	6411(11)
3.21	6411
31.11	6222
11.31	4422(2, 11)
21.21	4422
22.11	4332
3.111	621111
221.1	43311
21.111	422211
11.211	332211
111.111	222222
2.1111	42111111
11.1111	22221111
1.11111	2211111111

7.16. **Type D_6, Rationally Smooth z.** The unique maximal quotient map to a product of symmetic groups is $\varphi_{12}\colon W \to W(A_5)$.

		$\neg(\triangle)$	$\neg(\pm)$	$\neg(+)$	φ_{12}-inflation
.6	[11]1				✓
1.5	93	✓		✓	
.51	9111				✓
2.4	75	✓	✓	✓	
11.4	7311(11)			✓	
1.41	7311	✓	✓	✓	
$3 \cdot +$	$66 \cdot +$				
$3 \cdot -$	$66 \cdot -$				
21.3	5511(11)				
2.31	5511				
.42	7221				✓
.33	5331(11)				✓
1.32	5331				
.411	711111				✓
2.22	4431				
11.31	5322			✓	
111.3	531111(11)			✓	
1.311	531111			✓	
$21 \cdot +$	$4422 \cdot +$				

$21 \cdot -$	$4422 \cdot -$		
2.211	441111		
.321	522111		✓
11.22	3333		
1.221	333111		
111.21	332211(11)		
11.211	332211	✓	
.3111	51111111		✓
.222	322221		✓
1111.2	33111111(11)		
1.2111	33111111	✓	
.2211	32211111		✓
$111 \cdot +$	$222222 \cdot +$		
$111 \cdot -$	$222222 \cdot -$		
11.1111	22221111		
.21111	3111111111		✓
1.11111	2211111111		
.111111	111111111111		✓

7.17. **Type E_6, Rationally Smooth z.** The unique nontrivial quotient map is the sign character $\varepsilon \colon W \to \{\pm 1\} \simeq W(A_1)$.

		$\neg(\Delta)$	$\neg(\pm)$	$\neg(+)$	ε-inflation
$\phi_{1,0}$	E_6				✓
$\phi_{6,1}$	$E_6(a_1)$	✓	✓	✓	
$\phi_{20,2}$	D_5				
$\phi_{15,5}$	$E_6(a_3)(11)$				
$\phi_{30,3}$	$E_6(a_3)$			✓	
$\phi_{15,4}$	A_5				
$\phi_{64,4}$	$D_5(a_1)$			✓	
$\phi_{60,5}$	$A_4 + A_1$				
$\phi_{24,6}$	D_4				
$\phi_{81,6}$	A_4				
$\phi_{20,10}$	$D_4(a_1)(111)$			✓	
$\phi_{90,8}$	$D_4(a_1)(21)$			✓	
$\phi_{80,7}$	$D_4(a_1)$			✓	
$\phi_{60,8}$	$A_3 + A_1$				
$\phi_{10,9}$	$2A_2 + A_1$				
$\phi_{81,10}$	A_3				
$\phi_{60,11}$	$A_2 + 2A_1$				
$\phi_{24,12}$	$2A_2$				
$\phi_{64,13}$	$A_2 + A_1$				
$\phi_{15,17}$	$A_2(11)$				
$\phi_{30,15}$	A_2			✓	
$\phi_{15,16}$	$3A_1$				
$\phi_{20,20}$	$2A_1$				
$\phi_{6,25}$	A_1				
$\phi_{1,36}$	1				✓

Appendix A. The Exotic Fourier Transform

For completeness, this appendix provides the definition of the pairing $\{-,-\}$ from the character theory of finite reductive groups [Lus84].

Throughout, $\mathbf{k}$ is the algebraic closure of a finite field $\mathbf{F}$. As in the body of the article, G is a connected reductive algebraic group over $\mathbf{k}$ with Weyl group W, such that the characteristic of $\mathbf{k}$ is a good prime for G [Car93, 28]. Let $F \colon G \to G$ be the Frobenius map corresponding to the split $\mathbf{F}$-structure on G.

For all $w \in W$, the Deligne–Lusztig variety X_w is defined by the pullback square:

$$\begin{tikzcd} O_w & X_w \arrow[l] \arrow[d] \\ \mathcal{B} \times \mathcal{B} \arrow[u, leftarrow] & \mathcal{B} \arrow[l, "\mathrm{Id} \times F"'] \end{tikzcd}$$

In other words, $X_w \subseteq \mathcal{B}$ is the subvariety of Borels B such that $B \xrightarrow{w} F(B)$. The finite reductive group G^F acts on X_w, and hence on its compactly-supported ℓ-adic étale cohomology $\mathrm{H}^i_c(X_w, \bar{\mathbf{Q}}_\ell)$. The *Deligne–Lusztig character* $R_w \colon G^F \to \bar{\mathbf{Q}}_\ell$ is the virtual character defined by

$$R_w(g) = \sum_i (-1)^i \operatorname{tr}(g \mid \mathrm{H}^i_c(X_w, \bar{\mathbf{Q}}_\ell)).$$

For any $\psi \in \mathrm{Irr}(W)$, the *almost-character* $R_\psi \colon G^F \to \bar{\mathbf{Q}}_\ell$ is defined by

$$R_\psi = \frac{1}{|W|} \sum_{w \in W} \psi(w) R_w.$$

As in Section 1, let e be the identity of W. Then X_e is the finite set $\mathcal{B}^F$, and R_e is the character of the principal series representation of G^F on $\mathrm{H}^0(\mathcal{B}^F, \bar{\mathbf{Q}}_\ell)$.

Let $q = |\mathbf{F}|$. Fix a square root $q^{1/2} \in \bar{\mathbf{Q}}_\ell^\times$. By a classical theorem of Iwahori, the specialized Hecke algebra $H_W(q) := H_W(\mathsf{v})|_{\mathsf{v} \to q^{1/2}}$ acts on $\mathrm{H}^0(\mathcal{B}^F, \bar{\mathbf{Q}}_\ell)$ by operators commuting with G^F. Namely, δ_w sends the indicator function at a Borel B to the indicator on the set of Borels B' such that $B \xrightarrow{w} B'$. Therefore, R_e can be promoted to a bitrace $R_e \colon G^F \times H_W(q) \to \bar{\mathbf{Q}}_\ell$. We have a decomposition

$$R_e = \sum_{\chi \in \mathrm{Irr}(W)} \rho_\chi \otimes \chi_{\mathsf{v}}|_{\mathsf{v} \to q^{1/2}},$$

where ρ_χ is an irreducible character of G^F called the *unipotent principal series character* associated with χ.

Let $(-,-)_{G^F}$ be the scalar product on class functions on G^F. We define the *(truncated) exotic Fourier transform* $\{-,-\}$ by setting

$$\{\chi, \psi\} = (\rho_\chi, R_\psi)_{G^F} \quad \text{for all } \chi, \psi.$$

It turns out that this, indeed, only depends on W, and not on $\mathbf{F}$ or G.

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